Overall Survival Prediction in Adult Myeloid Leukemia: Kaplan–Meier, Cox Regression and Machine-Learning Survival Models

Bernard Frimpong
M.S. Applied Mathematics, University of Akron

October 2026

Abstract

This report documents my work on the QRT × Institut Gustave Roussy data challenge on Challenge Data (ENS Paris), whose goal is to rank adult patients with myeloid leukemia by their risk of death, using clinical measurements, cytogenetic reports and somatic gene mutations. Models are scored with the inverse probability of censoring weighted (IPCW) concordance index truncated at seven years. The data audit shows that the 1,193 test patients all come from a single clinical centre that is absent from the 23 training centres, and that this centre's patients are clinically sicker, so model validation will hold out whole centres. Kaplan–Meier analysis gives a median overall survival of 3.15 years, with strong separation by bone-marrow blasts, *TP53* and *SF3B1* mutations and monosomy 7 (log-rank $p < 0.001$ for all). The Kaplan–Meier estimator, a log-rank test and both concordance indices were computed by hand and match standard software. Single raw variables reach an IPCW C-index of at most 0.616. I then built 91 features from blood counts, a forgiving parser of the free-text karyotypes, and somatic mutations restricted to the 64 genes sequenced in both centres. The published IPSS-R clinical risk score, which needs no fitting, reaches an IPCW C-index of 0.669 ± 0.029 across 15 centre-grouped validation folds. Cox models evaluated on the same folds improve on it: a 24-variable model built from the IPSS-M gene list reaches 0.716 and an elastic-net Cox model on 84 features 0.722, each beating the IPSS-R on all 15 folds. Monocyte counts, missing in 74% of test patients, add nothing. Hazard ratios agree with clinical knowledge (TP53 multi-hit HR 3.2), and the proportional-hazards assumption is violated mainly by *NPM1* and platelets, whose effects change over time. Tree models add little: a random survival forest reaches 0.719, and XGBoost with a Cox loss, tuned by nested cross-validation and averaged over five seeds, reaches 0.724, a small but consistent gain (14 of 15 folds). A rank average of the elastic-net and XGBoost reaches 0.725 and beats the elastic-net on all 15 folds. Chosen as the final model before its public score was known, it scored 0.7552 on the unseen test centre, the best of all submissions, while XGBoost alone fell to 0.7504. Permutation importance on held-out centres shows that the mutation burden, counted on the gene panel shared by both centres, is the strongest signal. Against the published IPSS-M score, computed with no fitting (cross-validated 0.720), no fitted model is statistically better (ensemble +0.005, 95% CI −0.004 to +0.013), and the recalibrated IPSS-M is slightly better calibrated. Domain knowledge, whether built into the features or into the IPSS-M, accounts for nearly all of the predictive performance.

1 Introduction

Myeloid leukemias are blood cancers in which abnormal immature cells (blasts) accumulate in the bone marrow. Estimating a patient's prognosis guides treatment: low-risk patients may receive supportive care, while high-risk patients may be prioritised for intensive treatment such as stem cell transplantation.

The outcome is *overall survival* (OS): the time from diagnosis to death or to the last follow-up. When a patient is still alive at the last follow-up, the true survival time is unknown; we only know it is longer than the observed time. Such observations are called *right-censored*, and handling them correctly is the purpose of survival analysis.

The project follows a staged plan: Kaplan–Meier estimation and log-rank tests, Cox proportional hazards regression, and machine-learning survival models (gradient-boosted survival and random survival forests), all compared under the challenge metric with honest cross-validation. A similar crowdsourced effort, the DREAM 9 AML Outcome Prediction Challenge, also scored survival models with a concordance index [1].

2 Data

The data come from 24 clinical centres and are split into the five files described in Table 1. The `ID` column links the files.

Table 1: The five challenge files.

File	Rows	Content
`clinical_train.csv`	3,323	one row per patient: centre, blood counts, karyotype
`clinical_test.csv`	1,193	same, for test patients
`molecular_train.csv`	10,935	one row per somatic mutation
`molecular_test.csv`	3,089	same, for test patients
`target_train.csv`	3,323	`OS_YEARS`, `OS_STATUS`

Clinical variables. `BM_BLAST` (bone-marrow blasts, %), `WBC` (white blood cells, G/L), `ANC` (absolute neutrophil count, G/L), `MONOCYTES` (G/L), `HB` (haemoglobin, g/dL), `PLT` (platelets, G/L) and `CYTOGENETICS`, a free-text karyotype written in ISCN notation (for example `46,xy` for a normal male karyotype).

Molecular variables. Each row is one somatic mutation, with its chromosomal position, reference and alternate bases, the gene, the protein change, the broad effect, the variant allele fraction (`VAF`, the share of sequencing reads carrying the mutation) and, not mentioned in the challenge description, the sequencing read depth (`DEPTH`).

3 The evaluation metric

For each patient i we observe a time T_i (`OS_YEARS`), an event indicator Δ_i (1 = death, 0 = censored), and the model gives a risk score η_i.

Harrell's C-index. A pair of patients (i, j) is *comparable* when we know who died first, that is, when $T_i < T_j$ and patient i died ($\Delta_i = 1$). The pair is *concordant* when the model agrees, $\eta_i > \eta_j$. Harrell's C-index [2] is

$$C = \frac{\text{number of concordant comparable pairs}}{\text{number of comparable pairs}}. \tag{1}$$

A value of 0.5 corresponds to random guessing and 1 to a perfect ranking. Pairs with equal risk scores count as one half.

IPCW C-index (Uno's C). Patients who die late are less likely to be observed dying, because many patients are censored before then. Harrell's C therefore over-represents early deaths. Uno *et al.* [3] correct this by weighting each comparable pair by $1/\hat{G}(T_i)^2$, where $\hat{G}(t)$ is the Kaplan–Meier estimate of the probability of *not yet being censored* at time t. With the truncation horizon $\tau = 7$ years used by the challenge,

$$C_\tau = \frac{\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau) \, \mathbf{1}(\eta_i > \eta_j)}{\sum_i \sum_j \Delta_i \, \hat{G}(T_i)^{-2} \, \mathbf{1}(T_i < T_j,\ T_i < \tau)}. \tag{2}$$

In plain language: a death at a time when few patients remain uncensored is rare to observe, so it counts more. Only the ordering of the scores η_i matters. The challenge computes C_τ with `scikit-survival` [4].

4 Data audit (Stage 1)

4.1 Linking and outcome

All IDs are unique, every mutation row belongs to a known patient, and no patient appears in both training and test sets. However, 150 training patients have no recorded outcome and are excluded from modelling. Table 2 summarises the remaining outcomes and Figure 1 shows the observed times.

Table 2: Outcome of the training patients.

Quantity	Value
Training patients	3,323
Without outcome (excluded)	150
With outcome	3,173
Deaths ($\Delta = 1$)	1,600 (50.4%)
Censored ($\Delta = 0$)	1,573 (49.6%)
Median observed time (years)	1.65
Maximum observed time (years)	22.04
Observed time exactly 0 (all censored)	53
Observed beyond the 7-year horizon	172

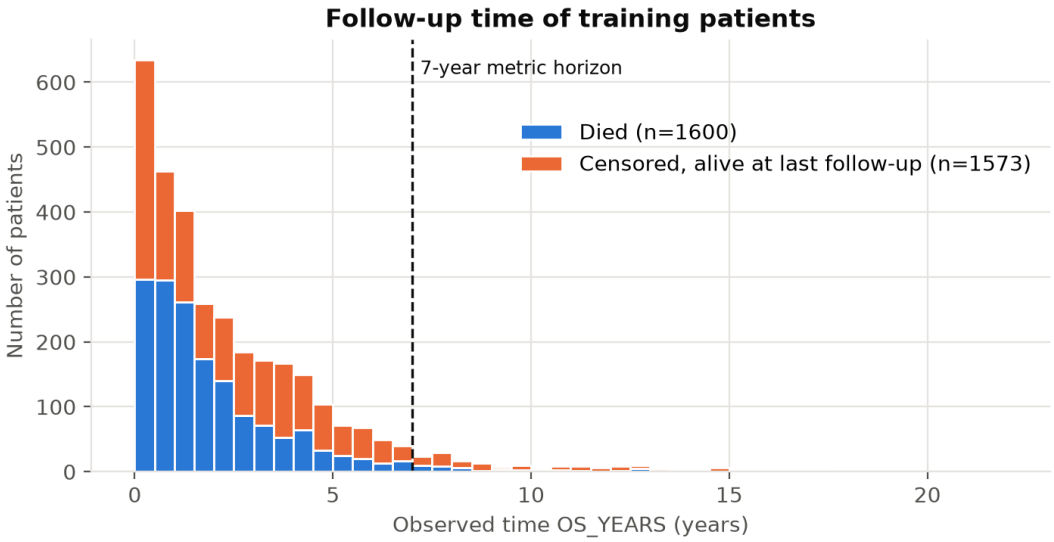


Figure 1: Observed time of the 3,173 training patients with an outcome, split into deaths and censored patients. The dashed line marks the 7-year horizon of the metric.

About half the patients are censored, so simply discarding them or treating their observed time as a death time would badly bias any model. Most deaths occur within the first two years, and only 172 patients are followed beyond seven years, which justifies the truncation of the metric.

4.2 Clinical centres: the test set is a new hospital

The 3,323 training patients come from 23 centres, while all 1,193 test patients come from one centre, `KYW`, that does not appear in training (Figure 2). Death rates also vary widely between training centres, from 11% (`ROM`) to 92% (`MUV`), reflecting different patient populations and follow-up lengths.

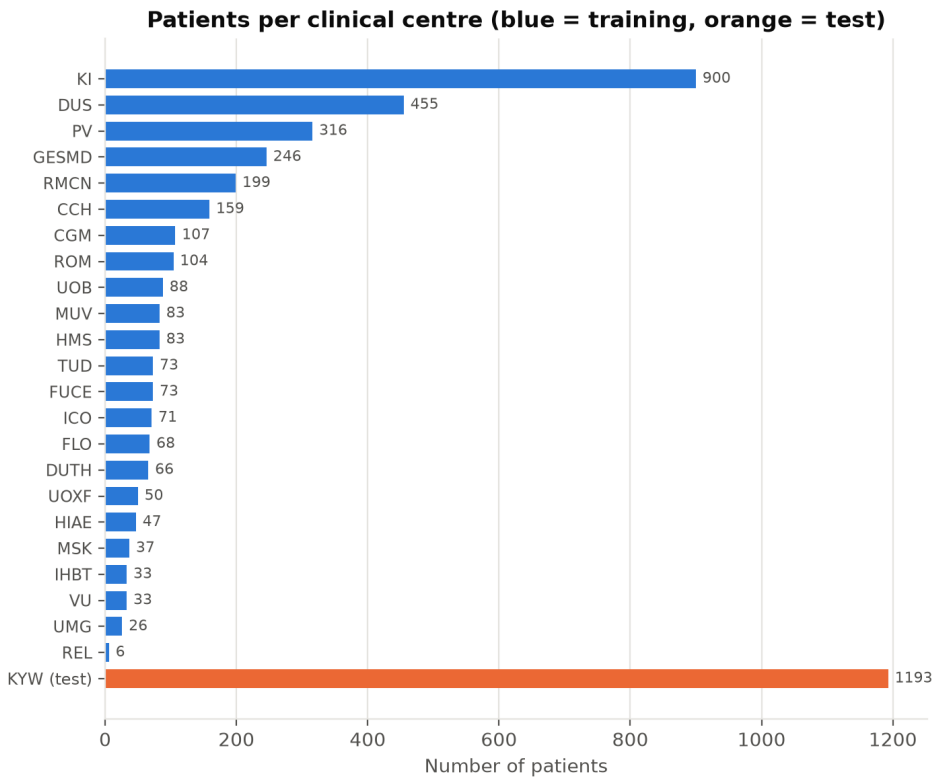


Figure 2: Number of patients per clinical centre. The test centre `KYW` (orange) contributes no training patients.

This has a direct consequence for validation. A random K-fold split mixes patients from the same centre into training and validation folds, so the model is always validated on centres it has already seen. The real task is to predict for an *unseen* centre. Cross-validation will therefore be grouped by centre, and the `CENTER` variable cannot be used as a predictor.

4.3 Clinical variables: training versus test

Table 3 and Figures 3 and 4 compare the clinical variables in the two sets.

Table 3: Missing values and medians of the clinical variables.

	Missing (%)		Median	
Variable	Train	Test	Train	Test
`BM_BLAST` (%)	3.3	9.6	3.00	6.00
`WBC` (G/L)	8.2	9.4	4.10	2.90
`ANC` (G/L)	5.8	11.8	2.00	1.30
`MONOCYTES` (G/L)	18.1	74.0	0.37	0.20
`HB` (g/dL)	3.3	9.3	9.70	8.40
`PLT` (G/L)	3.7	9.6	123.00	74.00
`CYTOGENETICS`	11.6	9.7	–	–

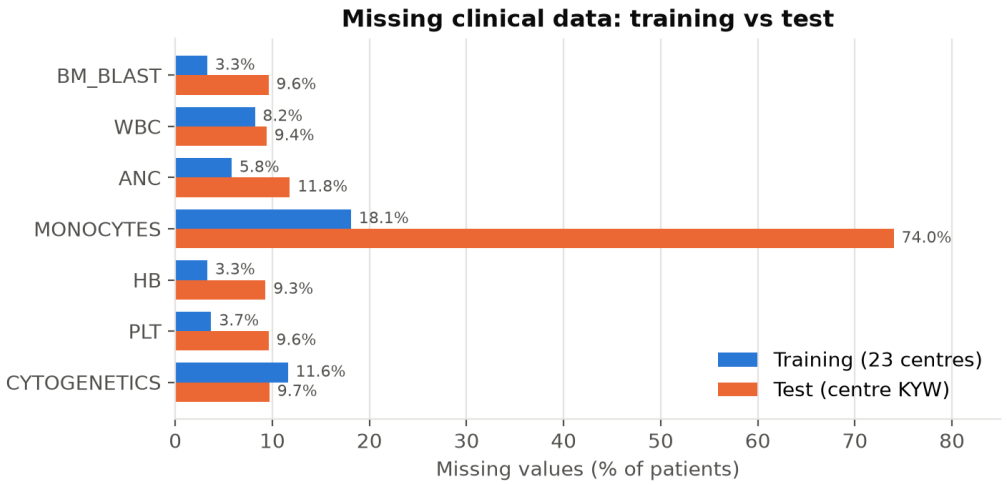


Figure 3: Percentage of missing values in each clinical variable.

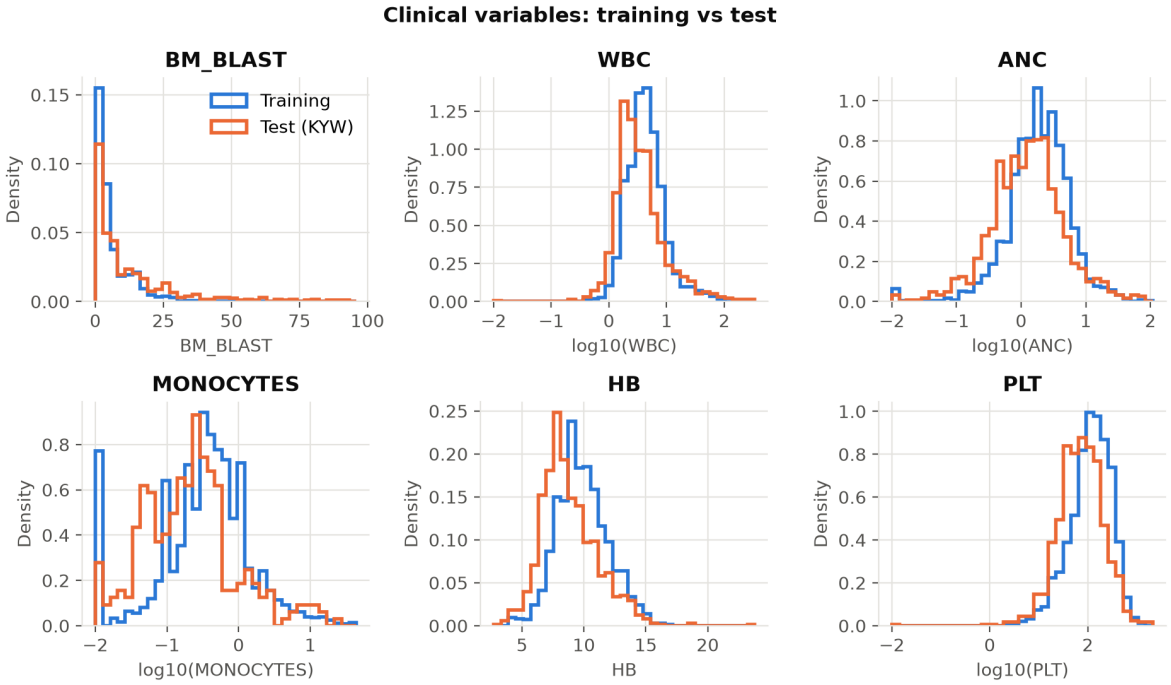


Figure 4: Distributions of the clinical variables in training (blue) and test (orange). Each histogram is scaled to area one so that sets of different sizes can be compared. Skewed blood counts are shown on a $\log_{10}$ scale; the spike at -2 for monocytes corresponds to recorded values of zero.

The test patients are clinically sicker: lower haemoglobin and platelets, lower neutrophils, and a much larger share with bone-marrow blasts above 20% (17.6% versus 4.2%). Monocytes are missing for 74% of test patients (883 of 1,193), so a model that leans on that variable would be fragile on the test set: all 883 would receive the same imputed value, and the training median (0.37) is above the test centre's own median (0.20).

4.4 Gene mutations

Training patients carry on average 3.29 recorded mutations and test patients 2.59; 297 training and 139 test patients have none, which is genuine information rather than missing data. Training data contain 124 distinct genes and test data 83, of which 19 never appear in training. Of the training rows, 114 lack a chromosomal position: they are partial tandem duplications of *MLL* (89) and internal tandem duplications of *FLT3* (25), which are structural events rather than single-base changes. In 1,274 cases the same gene is mutated more than once in one patient.

Table 4: Ten most frequently mutated genes in training: percentage of patients with at least one mutation.

Gene	Train (%)	Test (%)
TET2	31.1	17.5
ASXL1	26.9	12.7
SF3B1	22.2	7.7
SRSF2	17.2	9.1
DNMT3A	16.1	11.1
RUNX1	13.9	13.7
TP53	11.4	25.7
STAG2	9.1	6.7
U2AF1	8.6	8.4
EZH2	6.5	6.7

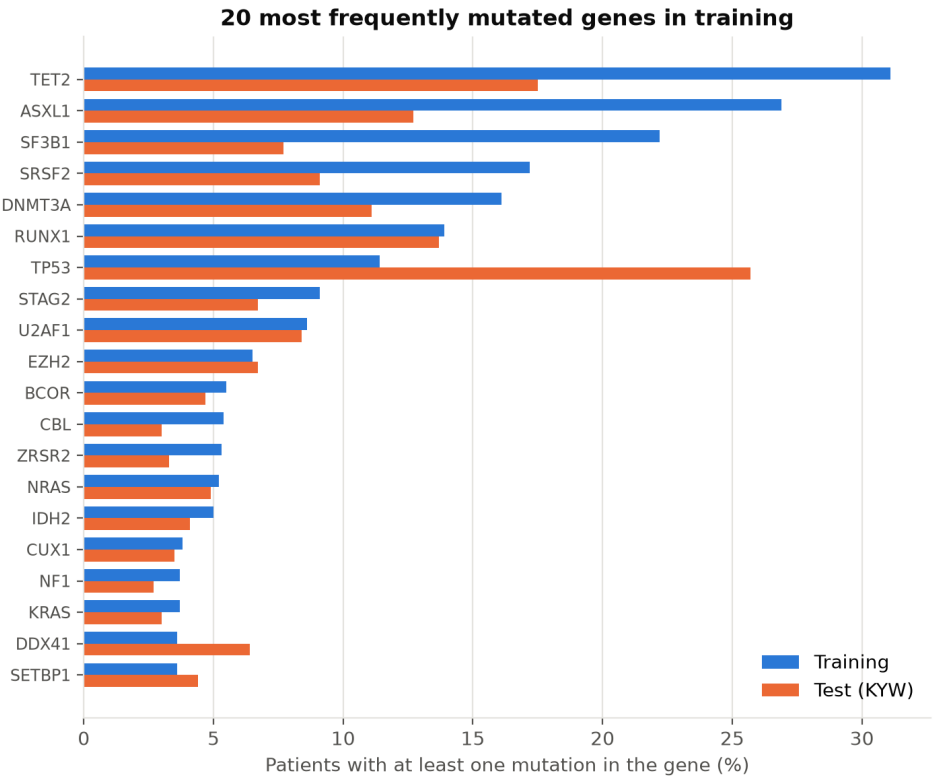


Figure 5: The twenty most frequently mutated genes in training, with their frequency in the test centre.

The most striking difference is *TP53*, mutated in 25.7% of test patients against 11.4% in

training (Table 4, Figure 5). *TP53* mutations are associated with poor prognosis in myeloid neoplasms, which is consistent with the sicker clinical profile of the test centre. By contrast, *SF3B1* mutations, typically associated with lower-risk disease, are almost three times rarer in the test set.

4.5 Cytogenetics text

In training, karyotypes are written in lower case (`46,xy`); in the test set they are written in upper case (`46,XY`), and 241 test patients have the single word `Normal`. Any parsing must first convert text to lower case and map `Normal` to a normal karyotype; otherwise every test karyotype would look unfamiliar to the model.

4.6 Consequences for modelling

1. Exclude the 150 training patients without outcome.
2. Validate with cross-validation grouped by centre, and do not use `CENTER` as a predictor.
3. Prefer features that are measured consistently in both sets; treat `MONOCYTES` with caution.
4. Normalise the cytogenetics text before extracting features.
5. Encode “no mutation found” as zero, not as missing.

5 Survival analysis without a model (Stage 2)

5.1 The Kaplan–Meier estimator

Let $t_1 < t_2 < \dots$ be the distinct times at which at least one death occurs. At each t_k let n_k be the number of patients still *at risk* just before t_k (neither dead nor censored yet) and d_k the number of deaths at t_k. The Kaplan–Meier estimator [5] of the survival function is

$$\hat{S}(t) = \prod_{k:\ t_k \le t} \left(1 - \frac{d_k}{n_k}\right). \tag{3}$$

In plain language: to survive past t_k you must survive every earlier death time, and at each one the chance of surviving, given you were still alive, is the share of the at-risk patients who did not die. Censored patients never make the curve drop; they simply leave the risk set, so they count fully for as long as they were observed.

Table 5 shows the first six steps. The first death occurs on day one ($t = 0.00274$ years) with $n = 3{,}120$ patients at risk: this is $3{,}173 - 53$, because the 53 patients with an observed time of zero were all censored at that time and left the risk set immediately. The first two steps are

$$\hat{S}(0.00274) = 1 - \tfrac{1}{3120} = 0.99968,$$
$$\hat{S}(0.00822) = 0.99968 \times \left(1 - \tfrac{2}{3114}\right) = 0.99968 \times 0.99936 = 0.99904.$$

My hand implementation of Equation (3) agrees with the `lifelines` library [7] at every death time to within 2×10^{-14}.

Table 5: First six steps of the Kaplan–Meier estimator, computed by hand.

t_k (years)	n_k	d_k	$1 - d_k/n_k$	$\hat{S}(t_k)$
0.00274	3120	1	0.99968	0.99968
0.00822	3114	2	0.99936	0.99904
0.01644	3107	1	0.99968	0.99872
0.02192	3103	2	0.99936	0.99807
0.02466	3098	1	0.99968	0.99775
0.02740	3095	2	0.99935	0.99711

The median overall survival, the first time at which $\hat{S}(t) \leq 0.5$, is **3.15 years**. Table 6 and Figure 6 give the whole curve.

Table 6: Kaplan–Meier overall survival with 95% confidence intervals.

Years	$\hat{S}(t)$	95% CI
1	0.790	0.774–0.804
2	0.614	0.595–0.633
3	0.509	0.489–0.529
5	0.364	0.342–0.386
7	0.275	0.250–0.299

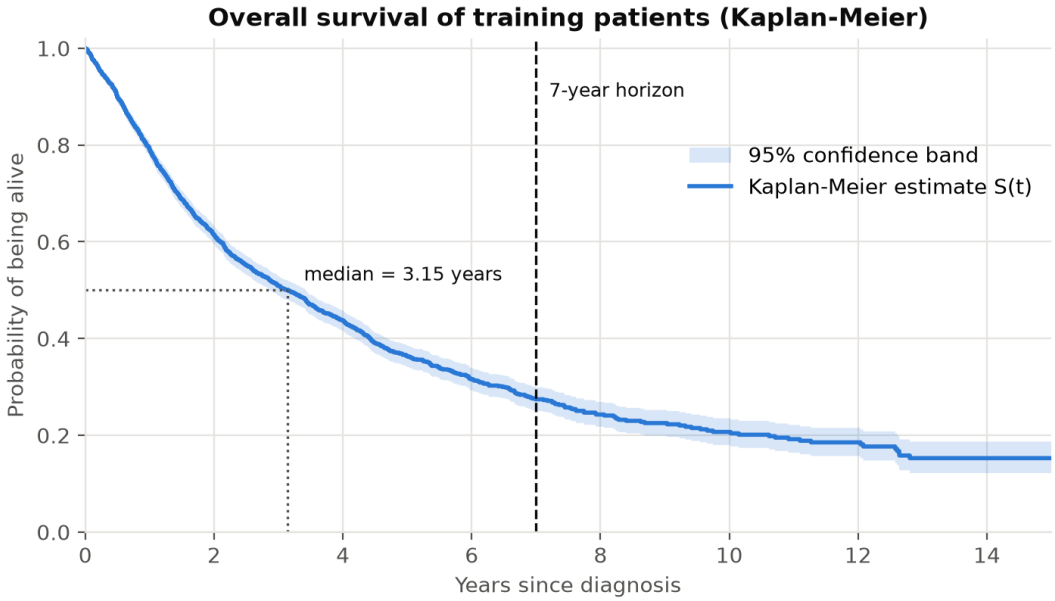


Figure 6: Kaplan–Meier estimate of overall survival for the 3,173 training patients, with a 95% confidence band.

5.2 Comparing groups: the log-rank test

To ask whether two groups have different survival, the log-rank test [6] compares, at each death time, the deaths observed in group 1 with the deaths it would receive if the groups were identical.

With n at risk in total, n_1 in group 1, $n_2 = n - n_1$, d deaths in total and d_1 in group 1,

$$E_{1k} = d\,\frac{n_1}{n}, \qquad V_k = \frac{n_1\, n_2\, d\,(n-d)}{n^2(n-1)}, \qquad \chi^2 = \frac{\left(\sum_k (d_{1k} - E_{1k})\right)^2}{\sum_k V_k}, \tag{4}$$

which follows a χ^2 distribution with one degree of freedom when the groups do not differ. In plain language: group 1 should get "its fair share" n_1/n of the deaths at each time; χ^2 measures how far the total gap is from zero, in units of its own variability.

Worked example: *TP53*. Summing over all death times, the 368 patients with a *TP53* mutation had $O_1 = 271$ observed deaths against $E_1 = 112.13$ expected, with total variance $V = 103.17$. Hence

$$\chi^2 = \frac{(271 - 112.13)^2}{103.17} = \frac{158.87^2}{103.17} = 244.64, \qquad p = 3.8 \times 10^{-55},$$

identical to the `lifelines` result. *TP53*-mutated patients died more than twice as often as expected under "no difference".

Table 7 and Figure 7 show four comparisons. Monosomy 7 was detected in the karyotype text as "`-7`" after a comma, slash or the start of the string and not followed by another digit; 79 patients without a blast value are left out of the blast comparison.

Table 7: Kaplan–Meier summaries and log-rank tests by group. All $p < 0.001$.

Comparison	Group	n	Deaths	Median (y)	$\hat{S}(3)$	χ^2 (df)
Blasts	$< 5\%$	2015	841	4.47	0.620	
	5–20%	942	622	1.68	0.326	272.1 (2)
	$> 20\%$	137	89	1.19	0.238	
TP53	not mutated	2805	1329	3.59	0.549	244.6 (1)
	mutated	368	271	1.00	0.207	
Monosomy 7	no	3008	1477	3.41	0.528	139.5 (1)
	yes	165	123	0.95	0.169	
SF3B1	not mutated	2446	1289	2.57	0.463	51.8 (1)
	mutated	727	311	4.56	0.665	

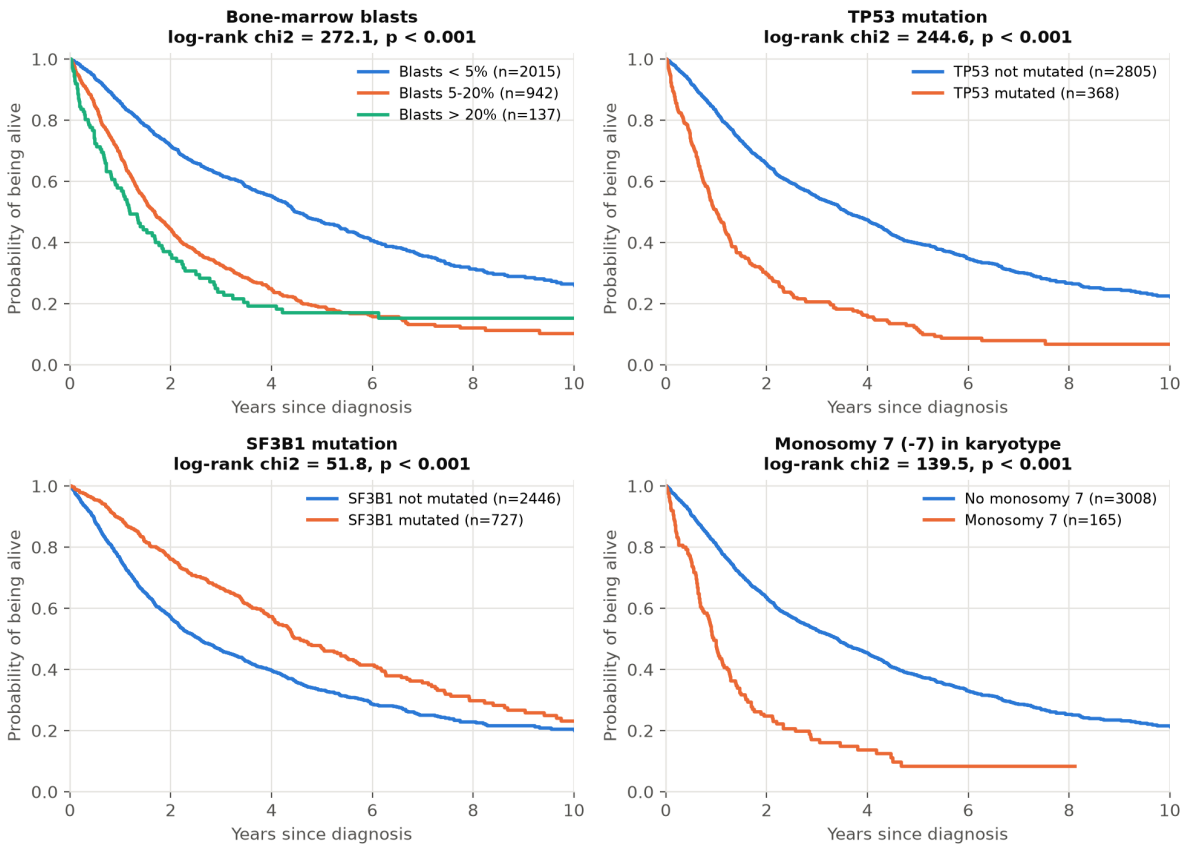


Figure 7: Kaplan–Meier curves by bone-marrow blast level, *TP53* mutation, *SF3B1* mutation and monosomy 7.

Higher blasts, *TP53* mutation and monosomy 7 all shorten survival sharply; the median survival of *TP53*-mutated patients is one year against 3.59 years for the rest. *SF3B1* is protective: mutated patients have a median survival of 4.56 years against 2.57. These directions agree with established clinical risk stratification and with the sicker profile of the test centre noted in Stage 1.

5.3 The censoring curve

Swapping the roles of death and censoring in Equation (3), so that the "event" is being censored, gives the *reverse Kaplan–Meier* estimate $\hat{G}(t)$: the probability that a patient is still being followed at time t. This is exactly the $\hat{G}$ in the metric (Equation (2)).

Table 8: Censoring curve and the resulting metric weight of a death at time t.

t (years)	$\hat{G}(t)$	weight $1/\hat{G}(t)^2$
1	0.827	$1/0.827^2 = 1.46$
3	0.613	$1/0.613^2 = 2.66$
5	0.347	$1/0.347^2 = 8.32$
7	0.198	$1/0.198^2 = 25.61$

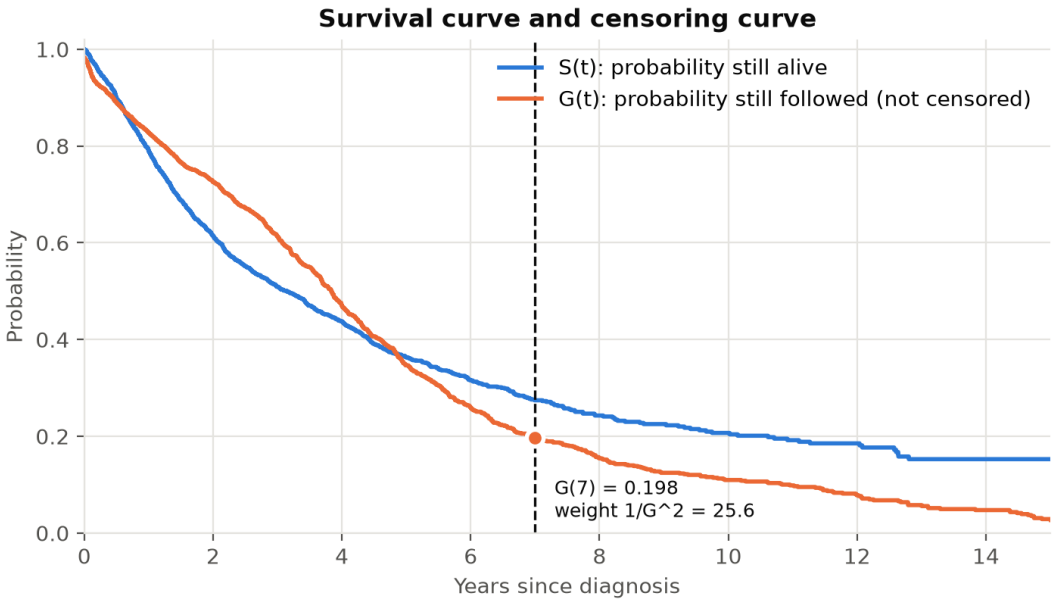


Figure 8: Survival curve $\hat{S}(t)$ and censoring curve $\hat{G}(t)$. By seven years only 19.8% of patients are still being followed.

Table 8 shows the practical effect: a pair started by a death just before seven years weighs about $25.6/1.46 \approx 17.5$ times as much as one started by a death at one year. A model that ranks late deaths poorly is therefore penalised far more by the challenge metric than by Harrell's C.

5.4 The metric by hand: a six-patient example

Figure 9 shows six patients with observed times, statuses and risk scores.

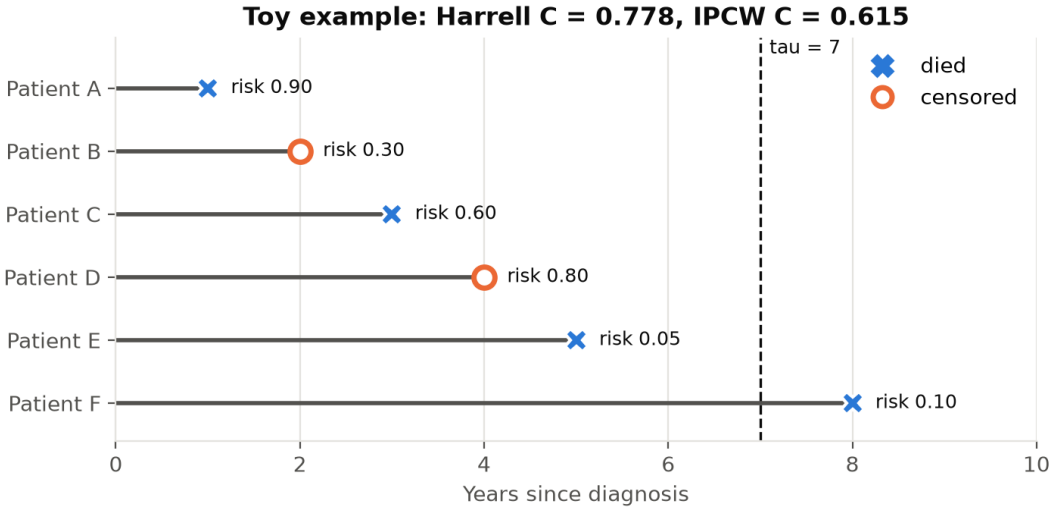


Figure 9: Toy example: crosses are deaths, circles are censored patients; each patient carries a risk score.

Comparable pairs. Only deaths can start a comparable pair, and only with patients observed for longer. Patient A (death at 1) is comparable with B, C, D, E, F; patient C (death at 3) with D, E, F; patient E (death at 5) with F; patient F (death at 8) has nobody after it. That gives

$5 + 3 + 1 = 9$ comparable pairs. Of these, two are discordant: (C, D) because $0.60 < 0.80$ and (E, F) because $0.05 < 0.10$. So

$$C_{\text{Harrell}} = \frac{7}{9} = 0.778.$$

Censoring curve. Censorings occur at $t = 2$ (B), with 5 patients at risk, and at $t = 4$ (D), with 3 at risk:

$$\hat{G}(2) = 1 - \tfrac{1}{5} = 0.8, \qquad \hat{G}(4) = 0.8 \times \left(1 - \tfrac{1}{3}\right) = 0.5333.$$

So $\hat{G}(1) = 1$, $\hat{G}(3) = 0.8$ and $\hat{G}(5) = 0.5333$, giving the weights

$$w_A = \frac{1}{1^2} = 1, \qquad w_C = \frac{1}{0.8^2} = 1.5625, \qquad w_E = \frac{1}{0.5333^2} = 3.5156.$$

Patient F died at $8 \geq \tau = 7$ and starts no pair anyway.

IPCW C-index.

$$\begin{aligned} \text{numerator} &= 5 \times 1 + 2 \times 1.5625 + 0 \times 3.5156 = 8.1250, \\ \text{denominator} &= 5 \times 1 + 3 \times 1.5625 + 1 \times 3.5156 = 13.2031, \\ C_7 &= \frac{8.1250}{13.2031} = 0.615. \end{aligned}$$

Both values match `scikit-survival` exactly. The IPCW value is lower because both wrong pairs start from later deaths (C at 3 years, E at 5 years), whose weights are larger.

5.5 Single-variable baselines

Before fitting any model, each variable can be used directly as a risk score, with its direction fixed by clinical knowledge rather than learned from the outcome (missing values set to the median). Table 9 and Figure 10 give the results on the 3,173 training patients.

Table 9: Concordance of single-variable risk scores ($\tau = 7$ years).

Risk score	Harrell's C	IPCW C
Random numbers (reference)	0.509	0.509
+ bone-marrow blasts	0.631	0.616
− platelets	0.631	0.608
+ number of mutations	0.602	0.605
− haemoglobin	0.603	0.593
+ *TP53* mutated (0/1)	0.566	0.553
− *SF3B1* mutated (0/1)	0.554	0.544
+ monosomy 7 (0/1)	0.532	0.526
− bone-marrow blasts (wrong sign)	0.369	0.384

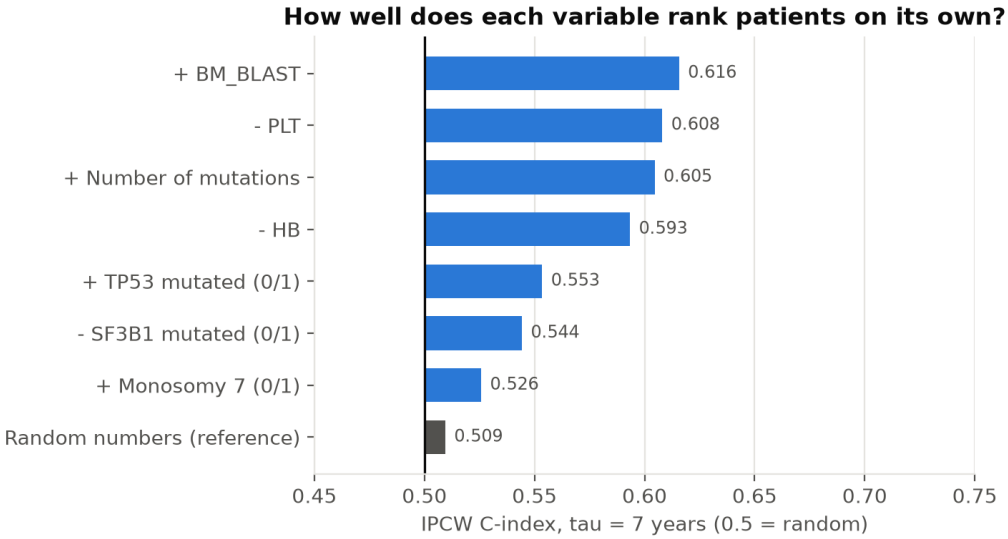


Figure 10: IPCW C-index of each single-variable risk score; 0.5 corresponds to random ranking.

Three lessons follow. First, reversing the sign of a score turns C into $1 - C$ ($0.631 \to 0.369$), so the direction of a risk score matters. Second, binary markers rank poorly on their own even when their log-rank test is overwhelming: *TP53* has $\chi^2 = 244.6$ but an IPCW C of only 0.553. Only 11.6% of patients carry the mutation, so most comparable pairs have equal scores and count one half. Statistical significance and ranking power are different things. Third, no single variable exceeds about 0.62, so combining them in a multivariable model is necessary.

6 Feature engineering (Stage 3)

Ninety-one features were built in five families (Table 10). Every feature is computed from one patient's own data, except two fixed choices that use only the predictors and never the outcome: the gene panel and the IPSS-R fill-in points. Missing clinical values are deliberately left empty at this stage, because the value used to fill them must be learned inside each cross-validation fold (Section 6.6).

Table 10: Feature families.

Family	Columns	Content
Clinical	6	blasts, haemoglobin, $\log(1+x)$ of 4 blood counts
Missing flags	6	1 if the clinical value was missing
Cytogenetic	18	abnormality count, 14 lesions, IPSS-R category, missing
Molecular summaries	15	counts, VAF, TP53, FLT3-ITD, MLL-PTD, 6 pathways
Gene indicators	45	1 if the gene is mutated
Clinical score	1	IPSS-R score
Total	91	

6.1 Clinical features

Blood counts are strongly right-skewed (Figure 4), so white cells, neutrophils, monocytes and platelets enter as $\log(1+x)$; the "1+" keeps recorded zeros finite. Blasts and haemoglobin are kept on their original scale. Each of the six variables also gets a 0/1 "was missing" flag.

6.2 Parsing the karyotype

A karyotype such as `46,xy,del(20)(q12)[2]/46,xy[18]` lists clones separated by "`/`", each with a chromosome count, the sex chromosomes, the abnormalities and a cell count in square brackets. Numerical changes are written `-7` (loss of chromosome 7) or `+8` (an extra chromosome 8); structural changes as `del`, `t`, `inv`, `der`, `add`, `i` and so on, and unidentified marker chromosomes as `mar`.

The text is irregular, especially in the test centre: upper and lower case, spaces, `{16}` or `[3/20]` cell counts, unbalanced brackets, and free text such as `Normal`, `complex` or `No mitosis`. The parser therefore searches for patterns instead of fully splitting the string:

1. lower-case the text, remove spaces, and convert `{}` to `[]`;
2. map free text: `Normal` to a normal karyotype, `complex` to at least three abnormalities, `>3 abnormalities` to four, and `No mitosis` or `Not evaluated` to missing;
3. delete cell counts, then collect the distinct numerical changes, the distinct structural changes and any marker chromosome across all clones;
4. do not count a translocation that only describes how a derivative chromosome was formed (`der(1)t(1;7)` is one abnormality), nor the common harmless variant `inv(9)(p12q13)`.

The parser was checked on 25 hand-written test strings covering these cases, and its monosomy 7 count (165 patients) agrees exactly with the simpler rule used in Stage 2. From the parsed karyotype the patient is assigned an IPSS-R cytogenetic category [9] (Table 11).

Table 11: IPSS-R cytogenetic categories as implemented.

Points	Category	Karyotype
0	Very good	−Y alone; del(11q) alone
1	Good	normal; del(5q), del(12p) or del(20q) alone; two including del(5q)
2	Intermediate	del(7q) alone, +8, +19, any other one or two abnormalities
3	Poor	−7; inv(3)/t(3;3)/del(3q); two including −7/del(7q); exactly three
4	Very poor	more than three abnormalities

Table 12 compares lesion frequencies. The test centre has nearly three times the training rate of complex karyotypes (31.3% against 10.9%) and of TP53 multi-hit status, defined as two or more TP53 mutations, a TP53 VAF of at least 0.5, or a TP53 mutation together with a 17p loss (18.9% against 6.7%).

Table 12: Lesion frequencies (% of patients with a usable karyotype; the first and last rows are % of all patients).

Lesion	Train	Test
Karyotype missing or unusable	11.0	10.1
Normal karyotype	58.2	38.2
Complex (≥ 3 abnormalities)	10.9	31.3
Monosomal karyotype	8.3	23.7
Monosomy 7	5.8	13.3
del(5q)	11.3	10.1
Trisomy 8	8.1	12.3
17p loss	2.1	7.4
TP53 multi-hit	6.7	18.9

6.3 Mutation features and the shared gene panel

The training file contains 124 genes and the test file 83, of which only 64 appear in both. Several genes frequent in training never appear in the test file, notably *KMT2C* (56 patients), *ZBTB33* (45), *EP300* (43) and *KMT2D* (32), which strongly suggests that the test centre used a smaller sequencing panel. If these genes were counted, "number of mutations" would mean something different in the two sets. All mutation features therefore use only the 64 shared genes; this removes 598 training mutations. A second mismatch was found: the test file never uses the label `splice_site_variant`, while 90 of its mutations have no effect label and are written in splice-site notation (`c.307+1G>A`); these were relabelled as splice-site variants.

The molecular features are: the number of mutations and of mutated genes, the maximum VAF, the number of truncating mutations (frameshift, stop-gained, splice-site), the number and maximum VAF of TP53 mutations, TP53 multi-hit status, FLT3 internal tandem duplication, MLL partial tandem duplication, six pathway counts (splicing, DNA methylation, chromatin, signalling, cohesin, transcription factors), and a 0/1 indicator for each of the 45 shared genes mutated in at least 1% (32) of training patients. Patients with no mutation receive zeros: "no mutation found" is information, not a missing value.

6.4 The IPSS-R score: a clinical benchmark

The revised International Prognostic Scoring System (IPSS-R) [8] is the standard clinical risk score for myelodysplastic syndromes, the dominant condition in this cohort (median blasts 3%). It adds points for five components:

Points	0	0.5	1	1.5	2–4
Cytogenetics	very good	–	good	–	2 int., 3 poor, 4 very poor
Blasts (%)	≤ 2	–	$(2, 5)$	–	2 if 5–10, 3 if > 10
Haemoglobin (g/dL)	≥ 10	–	8–<10	< 8	–
Platelets (G/L)	≥ 100	50–<100	< 50	–	–
ANC (G/L)	≥ 0.8	< 0.8	–	–	–

For example, a patient with a good karyotype (1 point), 7% blasts (2), haemoglobin 9.1 g/dL (1), platelets 45 G/L (1) and ANC 1.2 G/L (0) scores $1 + 2 + 1 + 1 + 0 = 5$, the "high" group. A missing component receives the median points of the training patients (1 for cytogenetics, blasts and haemoglobin; 0 for platelets and ANC). This uses no outcome information. Groups are very low (≤ 1.5), low $(1.5, 3]$, intermediate $(3, 4.5]$, high $(4.5, 6]$ and very high (> 6).

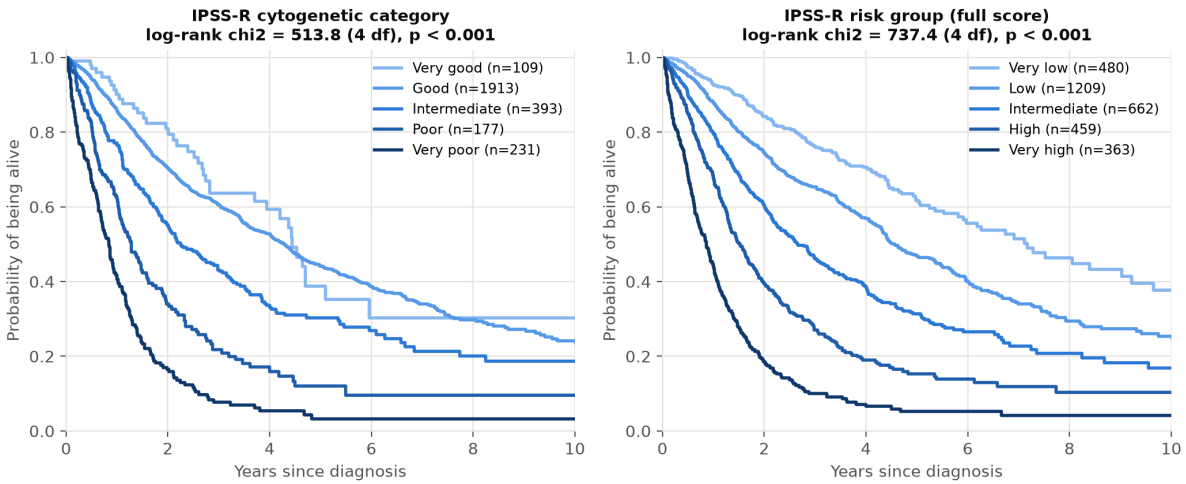


Figure 11: Kaplan–Meier curves by IPSS-R cytogenetic category (left) and by IPSS-R risk group (right) in the training data. Darker blue means higher risk.

Table 13: Survival by IPSS-R group in the training data.

Grouping	Group	n	Median (y)	$\hat{S}(3)$
Cytogenetic category ($\chi^2 = 513.8$)	Very good	109	4.44	0.636
	Good	1913	4.28	0.606
	Intermediate	393	2.26	0.432
	Poor	177	1.28	0.217
	Very poor	231	0.84	0.076
Risk group ($\chi^2 = 737.4$)	Very low	480	7.18	0.762
	Low	1209	4.56	0.654
	Intermediate	662	2.65	0.462
	High	459	1.52	0.270
	Very high	363	0.86	0.101

Both groupings separate survival cleanly and in the expected order (Figure 11, Table 13). The full score separates better than cytogenetics alone (χ^2 737.4 against 513.8), confirming that blasts and blood counts add information. Figure 12 applies the same score to the test centre: 33% of its patients are very high risk against 11% in training, and the median score is 5.0 against 3.0.

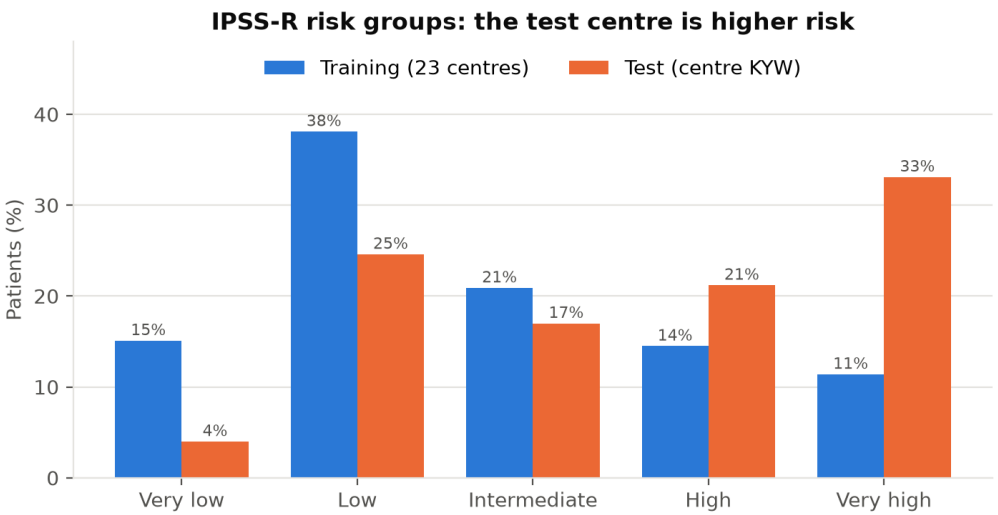


Figure 12: Share of patients in each IPSS-R risk group, training versus test.

6.5 Single-feature concordance

Figure 13 ranks features by their IPCW C-index when used alone (missing values filled with the median for this exploratory look only). The IPSS-R score is the strongest at 0.668, ahead of blasts (0.616) and the number of mutated genes (0.613). Platelets, haemoglobin, a normal karyotype and *SF3B1* mutation are protective ($C < 0.5$). Composite features such as counts and pathway sums rank better than single genes, for the tie-breaking reason found in Stage 2.

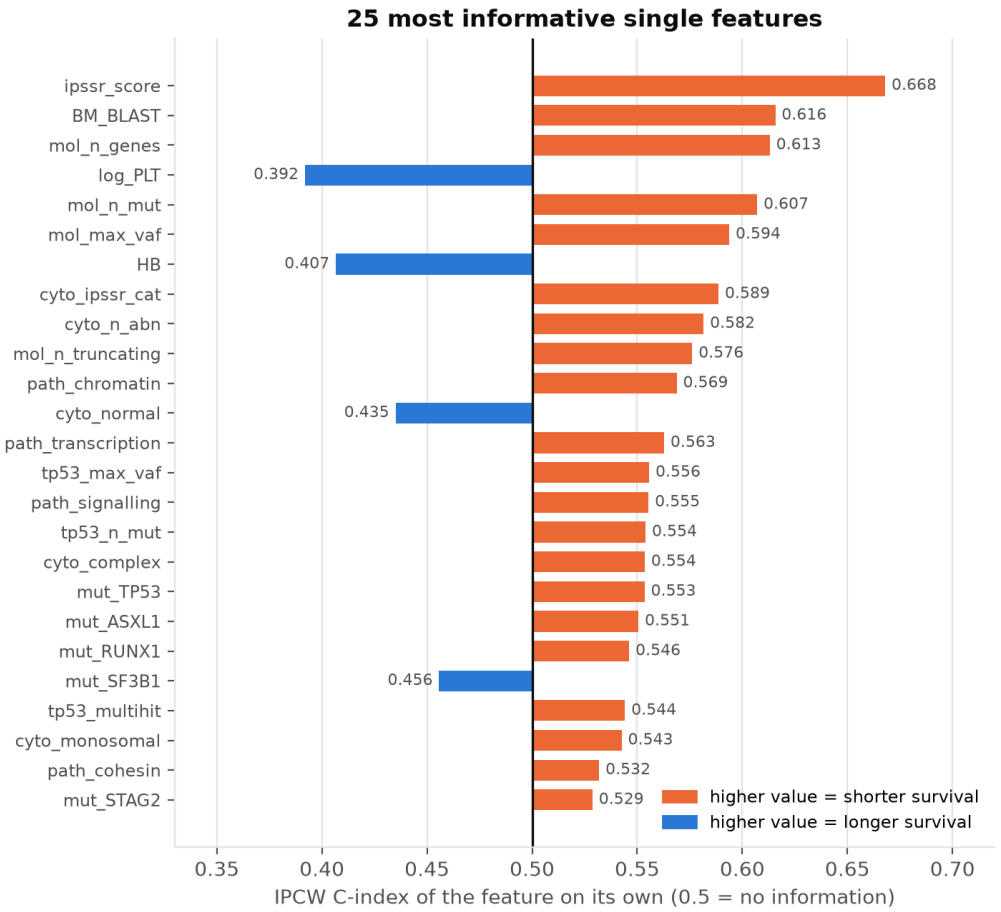


Figure 13: IPCW C-index of the 25 most informative features used alone. Orange: higher values go with shorter survival; blue: with longer survival.

6.6 Centre-grouped cross-validation

All patients of one centre are placed in the same fold, so each validation fold contains only centres unseen during training. The 23 centres are dealt into $K = 5$ folds like cards: they are put in a random order, and each in turn goes to the fold that currently has the fewest patients, which keeps the folds of similar size. Because there are only 23 centres, the split is repeated three times with different random orders, giving 15 validation folds. Two checks run in the code: no centre is split across folds, and the three repetitions are genuinely different partitions of the centres.

A bug found and fixed. The first version used `scikit-learn`'s `StratifiedGroupKFold` with `shuffle=True` and a different seed per repetition. The plot of fold scores in Stage 4 showed every value exactly three times: with only 23 groups, that splitter sorts the groups by their outcome profile before shuffling has any effect, so all three "repetitions" were the same five folds. The dealing procedure above replaced it, and the second check now guards against a repeat of the problem.

Table 14: Folds of the first repetition and the IPCW C-index of the (unfitted) IPSS-R score on each validation fold.

Fold	Patients	Centres	Death rate	Centre names	IPSS-R C_7
0	476	5	0.45	DUTH ICO MSK RMCN ROM	0.675
1	893	2	0.61	FUCE KI	0.659
2	478	6	0.38	CCH CGM FLO HIAE UOB VU	0.729
3	747	4	0.58	DUS GESMD HMS REL	0.629
4	579	6	0.40	IHBT MUV PV TUD UMG UOXF	0.679

The IPSS-R score is not fitted, so its variation across folds measures pure centre-to-centre noise. Over all 15 folds its IPCW C-index is 0.669 ± 0.029 (range 0.624–0.729), against 0.668 on all patients together. A difference in mean C-index of less than about 0.03 between two models is therefore not convincing by itself; the fair comparison is fold by fold (Section 7).

Filling missing values inside the folds. The median used to fill a missing value must be computed on the training part of each fold only. Otherwise the validation patients would help choose the fill-in value, a small leak of information from the data used to judge the model. Table 15 shows that these medians differ slightly between folds; Stage 4 will learn them inside each fold with a `scikit-learn` pipeline.

Table 15: Medians computed on the training part of each fold (repetition 0).

Fold	Blasts	Haemoglobin	$\log(1 + \text{PLT})$	$\log(1 + \text{monocytes})$
0	3.0	9.70	4.796	0.316
1	3.0	9.60	4.828	0.300
2	3.0	9.78	4.836	0.329
3	3.0	9.70	4.796	0.313
4	3.0	9.60	4.844	0.301
All patients	3.0	9.70	4.820	0.315

7 Cox regression (Stage 4)

7.1 The Cox proportional hazards model

The *hazard* $h(t)$ is the instantaneous risk of death at time t for someone still alive at t. The Cox model [10] writes the hazard of a patient with features $x = (x_1, \dots, x_p)$ as

$$h(t \mid x) = h_0(t)\, \exp(\beta_1 x_1 + \cdots + \beta_p x_p)\,, \tag{5}$$

where $h_0(t)$ is a baseline hazard shared by everyone and left completely unspecified. In plain language: every patient's risk curve has the same shape over time, scaled up or down by a patient-specific factor $\exp(\beta^\top x)$. The ratio of two patients' hazards therefore does not depend on t (the *proportional hazards* assumption), and e^{β_k} is the *hazard ratio* (HR) for a one-unit increase in x_k with the other features held fixed. The linear predictor $\eta = \beta^\top x$ is exactly the risk score the challenge asks for.

The coefficients are found by maximising the *partial likelihood*

$$L(\beta) = \prod_{i:\ \Delta_i = 1} \frac{\exp(\beta^\top x_i)}{\sum_{j:\ T_j \geq T_i} \exp(\beta^\top x_j)}. \tag{6}$$

Each factor asks: of all patients still at risk when patient i died, how likely was it to be patient i? Censored patients appear in the denominators while they are at risk, which is how the model uses them, and $h_0(t)$ cancels out.

Worked example. TP53 multi-hit has HR 3.21 in the model below, so its coefficient is $\ln 3.21 = 1.166$. Haemoglobin has HR 0.768 per standard deviation (2.0 g/dL), so $\beta = \ln 0.768 = -0.264$. A patient with TP53 multi-hit whose haemoglobin is one standard deviation *below* average, all else equal, has

$$\mathrm{HR} = \exp(1.166 - (-0.264)) = \exp(1.430) = 4.18 = 3.21 \times \frac{1}{0.768},$$

that is, about four times the instantaneous risk of death of an otherwise average patient.

7.2 Penalised Cox: the elastic net

With 84 features, some rare, ordinary maximisation of Equation (6) over-fits. The elastic net [11, 12] instead maximises

$$\ln L(\beta) \; - \; \alpha \left(\rho \sum_k |\beta_k| + \frac{1-\rho}{2} \sum_k \beta_k^2 \right), \tag{7}$$

where $\alpha > 0$ sets the overall strength of the penalty and $\rho \in (0, 1]$ the mix: $\rho = 1$ is the lasso, which sets many coefficients exactly to zero, and small ρ is close to ridge regression, which shrinks all coefficients towards zero without removing them. Features are standardised first so the penalty treats them equally.

Nested cross-validation. α and ρ must be chosen without looking at the validation fold. Inside each outer training set, an inner four-fold split, also grouped by centre, scores 30 values of α for each $\rho \in \{0.1, 0.5, 1\}$; the best pair is refitted on the whole outer training set and only then applied to the outer validation fold. Missing values are filled with medians and features are standardised inside a `scikit-learn` pipeline, so these steps are also learned only on the data the model is trained on.

7.3 Models compared

All models are evaluated on the same 15 centre-grouped validation folds, with the censoring curve $\hat{G}$ of the metric estimated on each fold's training part:

A. IPSS-R score the published score, no fitting (reference).

B. Benchmark-style Cox blasts, haemoglobin, platelets and the number of mutations. The challenge's benchmark notebook is available only to registered participants; this is my reconstruction from the challenge description and may differ in detail.

C. IPSS-M-style Cox the five blood values, the IPSS-R cytogenetic category, a flag for a missing karyotype, the 16 "main effect" genes of the IPSS-M molecular score [18] (including TP53 multi-hit and MLL-PTD), and the number of mutated IPSS-M "residual" genes capped at 2. These 24 variables come from the literature, not from this data. A very small ridge penalty (0.1) keeps the fit stable.

D. Elastic-net Cox all 84 features except monocytes and the missing flags.

E. Elastic-net Cox + D plus log monocytes and the six missing flags.

7.4 Cross-validated results

Table 16: IPCW C-index ($\tau = 7$) over 15 centre-grouped validation folds. "Paired" is the mean over folds of (model C − IPSS-R C on the same fold).

Model	Mean	SD	Range	Harrell's C	Paired	Folds won
A. IPSS-R score	0.669	0.029	0.624–0.729	0.692	–	–
B. Benchmark-style Cox	0.688	0.029	0.657–0.756	0.703	+0.019	13/15
C. IPSS-M-style Cox	0.716	0.023	0.686–0.765	0.736	+0.047	15/15
D. Elastic-net Cox	**0.722**	0.021	0.696–0.769	0.740	+0.053	15/15
E. Elastic-net Cox +	0.721	0.021	0.696–0.769	0.739	+0.053	15/15

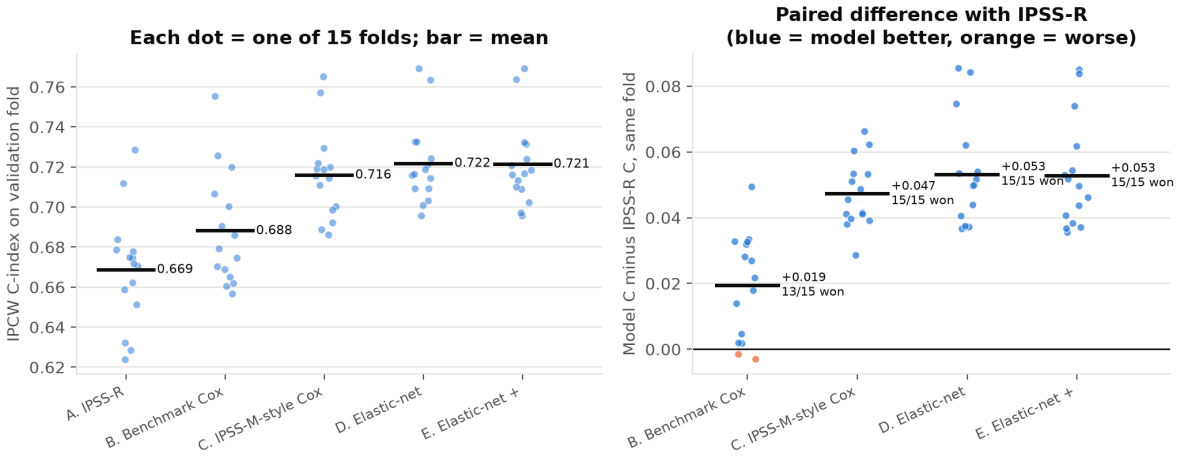


Figure 14: Left: IPCW C-index of each model on each of the 15 validation folds. Right: the paired difference with the IPSS-R score on the same fold.

Table 17: Further paired comparisons over the 15 folds.

Comparison	Question	Mean diff.	SD	Folds won
C − B	clinical feature set vs benchmark	+0.028	0.014	14/15
D − C	84 penalised features vs 24 chosen	+0.006	0.009	10/15
E − D	adding monocytes and missing flags	−0.0004	0.001	4/15

Four conclusions follow from Tables 16 and 17 and Figure 14:

1. Every fitted Cox model beats the unfitted IPSS-R. The gains of C, D and E are convincing: about +0.05, larger than the fold-to-fold spread, and won on all 15 folds.

2. The real step forward is the feature set. Moving from the four benchmark variables to the 24 literature-chosen variables adds +0.028 and wins 14 of 15 folds.

3. Letting a penalised model use all 84 features adds only +0.006 beyond those 24 variables (10 of 15 folds), a small and uncertain gain.

4. Monocytes and the missing-value flags add nothing and lose 11 of 15 folds. This settles the Stage 1 concern: the variable missing for 74% of test patients is safely dropped.

On the training data itself, the IPSS-M-style model reaches a concordance of 0.743 against 0.716 in cross-validation: the gap is the optimism of judging a model on the patients it was fitted to.

7.5 Hazard ratios

Figure 15 shows the hazard ratios of the IPSS-M-style model refitted on all 3,173 patients (missing values filled with the median; blood values per standard deviation, the cytogenetic category and residual-gene count per step, and genes as mutated versus not).

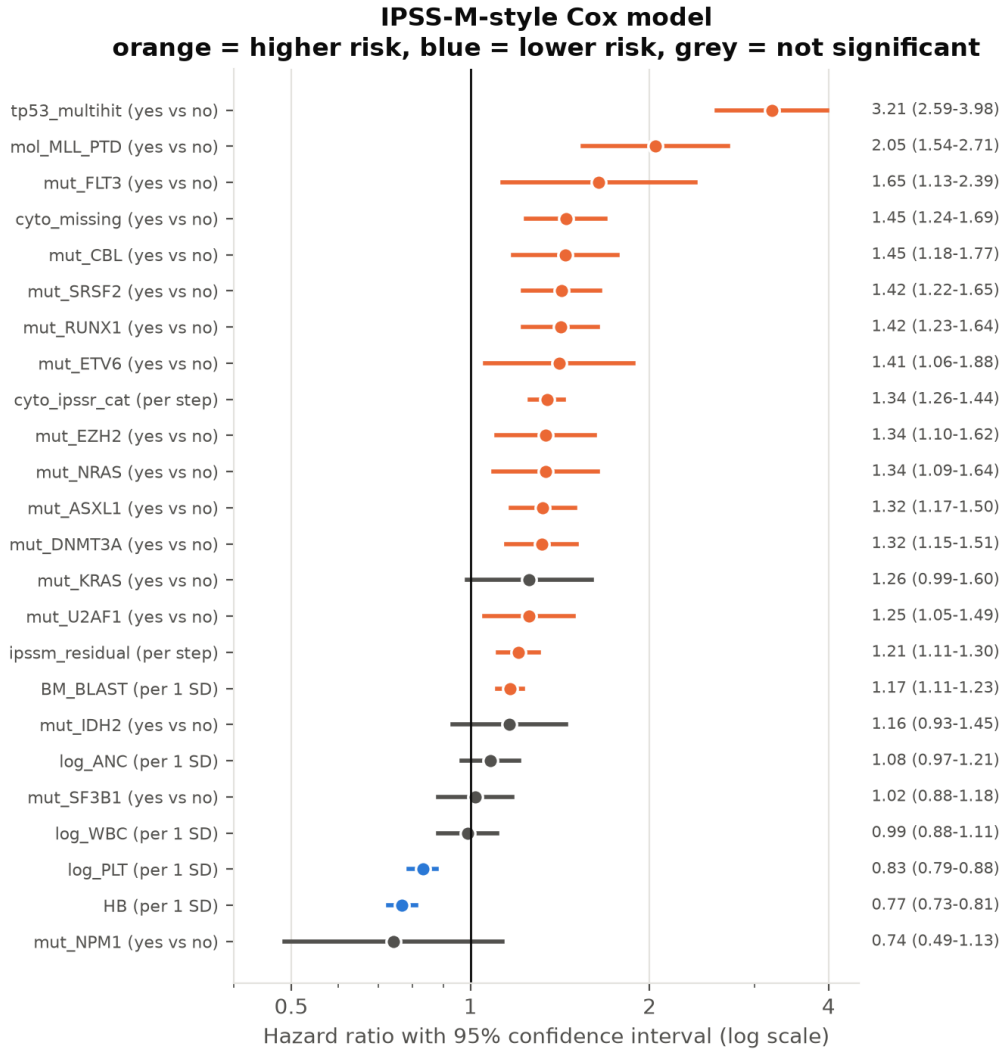


Figure 15: Hazard ratios with 95% confidence intervals of the IPSS-M-style Cox model. Orange: significantly higher risk; blue: significantly lower risk; grey: interval includes 1.

TP53 multi-hit is by far the strongest factor (HR 3.21, 95% CI 2.59–3.98), followed by MLL partial tandem duplication (2.05) and FLT3 mutation (1.65). Each step up the IPSS-R cytogenetic scale multiplies the hazard by 1.34, and a missing karyotype by 1.45, perhaps because failed karyotypes are more common in sicker patients. Higher haemoglobin (HR 0.77 per SD) and platelets (0.83 per SD) are protective. Two results differ from Stage 2. First, *SF3B1* is no longer protective (HR 1.02, CI 0.88–1.18): its favourable Kaplan–Meier curve is explained by the other variables, since *SF3B1*-mutated patients tend to have low blasts and a good karyotype. Second, white cells and neutrophils add nothing once the other variables are known.

7.6 Checking proportional hazards

If the proportional-hazards assumption holds, each coefficient β_k is constant over time. Grambsch and Therneau [13] showed that the scaled Schoenfeld residual of a death at time t, plus $\hat{\beta}_k$, estimates $\beta_k(t)$; a trend over time signals a violation. Table 18 gives the test for the largest statistics.

Table 18: Proportional-hazards test (rank-transformed time), largest statistics.

Variable	Test statistic	p
NPM1 mutated	23.10	1.5×10^{-6}
Karyotype missing	17.07	3.6×10^{-5}
log platelets	16.04	6.2×10^{-5}
SF3B1 mutated	5.12	0.024
U2AF1 mutated	3.64	0.056

Four of 24 variables have $p < 0.05$. With 1,600 deaths even small departures become significant, so Figure 16 shows the size of the two clearest ones. *NPM1* is harmful in the first months ($\beta(t) \approx +2$) and strongly protective afterwards ($\beta(t) \approx -2.5$ beyond four years): its single average coefficient of -0.30 hides two opposite effects. The protection of a high platelet count is concentrated in the first year and fades to almost nothing after three years. The other 20 variables show no evidence of change. Because the metric only ranks patients, mild violations matter less than they would for predicting survival probabilities, but they motivate the tree-based models of Stage 5, which do not assume proportional hazards.

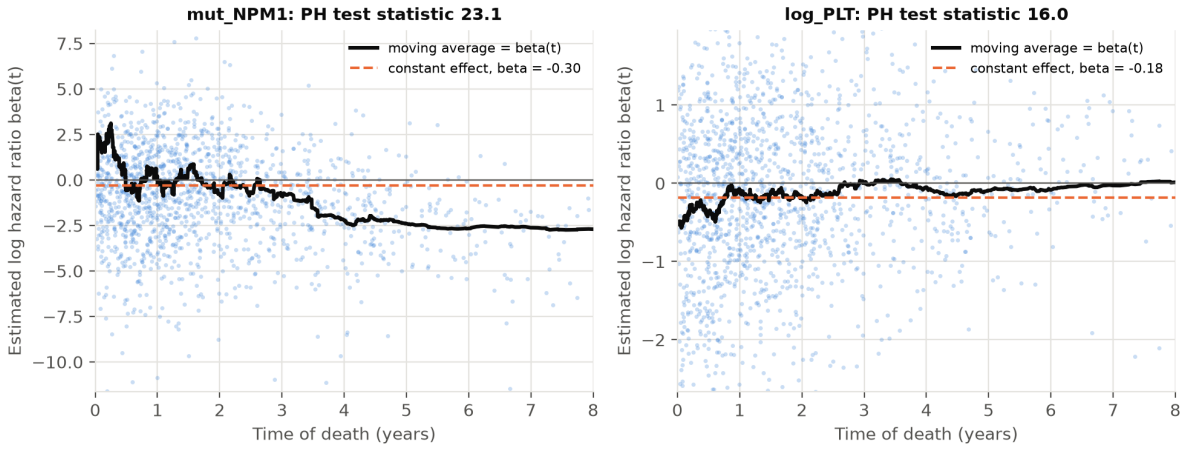


Figure 16: Estimated time-varying effect $\beta(t)$ (scaled Schoenfeld residual plus $\hat{\beta}$; black line: moving average of 150 deaths) against the constant effect assumed by the Cox model (dashed).

7.7 The final elastic-net model

Refitted on all training patients with the same nested procedure (inner five-fold centre-grouped split), the elastic net chooses $\rho = 0.1$, close to ridge, and $\alpha = 0.190$, keeping 44 of 84 coefficients non-zero (Figure 17). The largest effects are the IPSS-R score itself, the number of mutated genes, the TP53 and overall maximum VAF, and, on the protective side, haemoglobin and platelets.

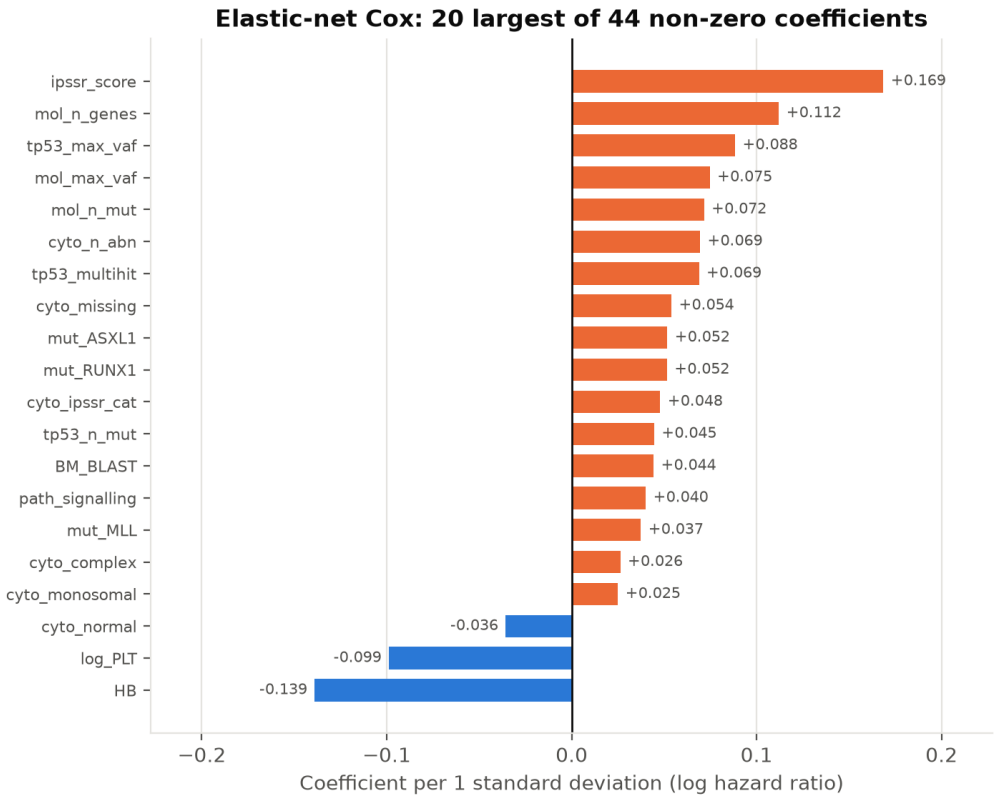


Figure 17: The 20 largest coefficients of the final elastic-net Cox model, per standard deviation of each feature.

To see what a C-index of 0.72 means in practice, each training patient was given the risk score of the elastic-net model trained *without their centre* (first repetition), and patients were split into quartiles within each fold. Figure 18 and Table 19 show the result.

Table 19: Survival by out-of-fold risk quartile (elastic-net Cox, unseen centres).

Quartile	n	Median (y)	$\hat{S}(3)$	$\hat{S}(7)$
Q1 (lowest risk)	795	8.28	0.834	0.564
Q2	793	4.40	0.643	0.326
Q3	791	2.42	0.431	0.157
Q4 (highest risk)	794	1.07	0.130	0.057

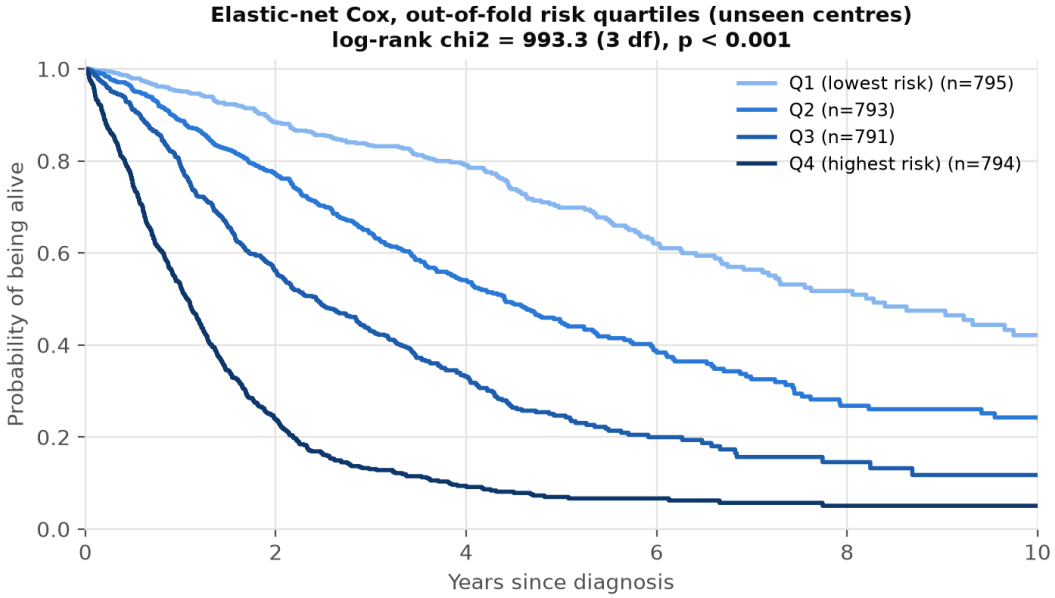


Figure 18: Kaplan–Meier curves by out-of-fold risk quartile; log-rank $\chi^2 = 993.3$ (3 df).

The median survival falls from 8.3 years in the lowest-risk quarter to 1.1 years in the highest, and three-year survival from 83% to 13%. For comparison, the five IPSS-R groups gave $\chi^2 = 737.4$ with four degrees of freedom (Stage 3).

7.8 Submissions

Three submission files were produced from models fitted on all 3,173 training patients: the benchmark-style Cox, the IPSS-M-style Cox and the elastic-net Cox. Each contains the 1,193 test IDs and a `risk_score`. As a sanity check, the elastic-net mean risk score is +0.43 on the test centre against 0 on the training data, consistent with the sicker test population. Two files were submitted (Table 20).

Table 20: Public leaderboard scores of the Stage 4 submissions (IPCW C-index on the public part of the test centre).

Model	Cross-validation	Public score
D. Elastic-net Cox (84 features)	0.722	**0.7543**
C. IPSS-M-style Cox (24 variables)	0.716	0.7374
Difference D − C	+0.006	+0.017

Two observations. First, both public scores are *higher* than cross-validation, the reverse of the usual drop on a new centre. A plausible reason is the case mix found in Stage 3: the test centre spreads its patients over a wider range of risk (a third are IPSS-R very high, yet low-risk patients remain), and a population with a wider range of risk is easier to rank, so the C-index rises. The leaderboard score therefore reflects the centre as much as the model, and models will continue to be chosen by paired cross-validation rather than by public score.

Second, the gap between the two models almost triples at the test centre, from +0.006 in cross-validation to +0.017. Both scores are measured on the same public patients, so the comparison is paired. A likely explanation is that the elastic-net's extra features describe the high-risk end in detail (number of abnormalities, monosomal karyotype, TP53 VAF, number of mutated genes), whereas model C summarises it coarsely (the IPSS-R cytogenetic category

stops at "more than three abnormalities"). The test centre has many more such patients than the training centres. A lesson for the write-up: cross-validation on centres unlike the target can understate the value of features that matter most for the target.

8 Machine-learning survival models (Stage 5)

Cox models assume that each feature acts the same way at all times (proportional hazards) and adds to the log hazard without interacting with the others. Tree-based models drop both assumptions. Two were fitted on the same 84 features as the elastic-net and evaluated on the same 15 centre-grouped folds.

8.1 Random survival forest

A random survival forest [14] grows many survival trees. Each tree is grown on a bootstrap sample of patients (drawn with replacement) and, at every split, considers only a random subset of the features. A split is chosen to separate the survival curves of the two child nodes as much as possible (by the log-rank statistic of Stage 2). Each leaf stores the Nelson–Aalen estimate of the cumulative hazard of its patients, and the forest averages these over trees. The risk score is the predicted cumulative hazard summed over the observed event times: more expected deaths means higher risk. Settings: 300 trees, at least 15 patients per leaf, and $\sqrt{84} \approx 9$ features tried per split. These were fixed rather than tuned, because forests are not very sensitive to them and one fit takes about six seconds.

8.2 Gradient boosting with a Cox loss

Gradient boosting [15] builds the risk score as a sum of small trees,

$$\eta(x) = \nu \sum_{m=1}^{M} f_m(x),$$

added one at a time. Each new tree f_m is fitted to the gradient of the negative Cox log partial likelihood (Equation (6)) at the current scores. In plain language, each tree pushes the scores in the direction that most improves the Cox fit, and the learning rate $\nu = 0.03$ keeps every step small. The XGBoost implementation [16] was used, with 80% of patients and half the features sampled for each tree and a minimum leaf weight of 10. It handles missing values itself by learning which branch they should follow. The tree depth (2, 3 or 4) and the number of trees (50 to 600) were chosen by the same nested, centre-grouped inner cross-validation as the elastic-net. In the first run, the smallest number of trees offered (100) was chosen in 7 of 15 folds. A choice at the edge of the grid means the best value might lie beyond it, so 50 and 150 were added. After that, 50 was never chosen, and the choices spread between 100 and 300 trees.

Computational reproducibility. The first version used XGBoost's fast approximate ("histogram") split finding with two threads. Run on Linux and on Windows with the same XGBoost version (3.4.1), every other model in this report gave identical results, but XGBoost did not: a cross-validated mean of 0.7227 on one machine and 0.7238 on the other, with different depths chosen in some folds. That difference (0.001) was as large as XGBoost's whole advantage over the elastic-net at the time. The final version uses exact split finding on a single thread and averages five models with different random seeds (42 to 46) at the depth and number of trees chosen by the inner cross-validation. Each machine now reproduces its own results exactly. A small difference between machines remains, because the random row and column subsampling relies on random-number routines of the C++ standard library, which the Windows and Linux

compilers implement differently. It shrank to 0.0002 in every summary (for example, the ensemble of Section 9 gains +0.0032 over the elastic-net on Windows and +0.0030 on Linux, winning 15 of 15 folds on both), and no conclusion depends on it. All numbers for XGBoost and the ensembles in this report are from the author's Windows run.

8.3 Results

Table 21: IPCW C-index over the same 15 centre-grouped folds; paired differences are with the elastic-net Cox (D).

Model	Mean	SD	Range	Paired vs D	Folds won vs D
A. IPSS-R score	0.669	0.029	0.624–0.729	−0.053	–
D. Elastic-net Cox	0.722	0.021	0.696–0.769	–	–
F. Random survival forest	0.719	0.021	0.694–0.774	−0.003	4/15
G. XGBoost Cox (5 seeds)	**0.724**	0.020	0.697–0.767	+0.0026	14/15

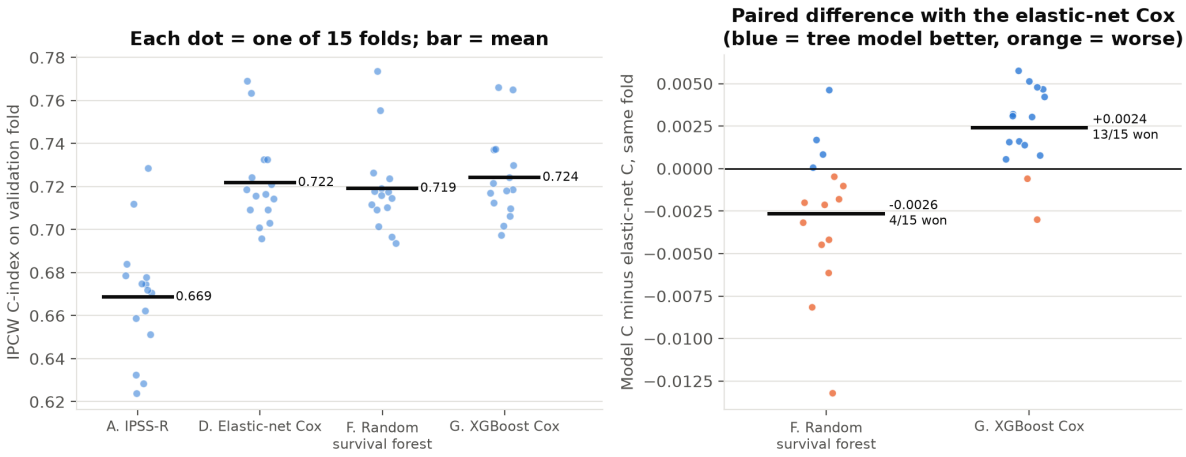


Figure 19: Left: IPCW C-index on each validation fold. Right: paired difference with the elastic-net Cox on the same fold.

The tree models bring at most a small gain over the elastic-net (Table 21, Figure 19). Seed-averaged XGBoost improves on it by +0.0026 (paired SD 0.0023), winning 14 of 15 folds: small, but consistent. It also beats the random survival forest by +0.0052 (14 of 15 folds), which is slightly worse than the elastic-net (−0.003, 4 of 15). The inner cross-validation consistently preferred small models (mostly 100 to 300 trees of depth 3 or 4 at a low learning rate), which signals that there is little structure left for trees to find beyond what the features already encode. Two reasons are likely. The engineered features already contain the main non-linearities and interactions, since the IPSS-R score, the cytogenetic categories and TP53 multi-hit status are themselves expert-designed combinations. And with about 1,600 deaths, the data are too small to estimate reliably the subtle time-varying effects found in Stage 4.

Accuracy by horizon. Figure 20 recomputes the out-of-fold IPCW C-index with horizons τ from one to seven years. All models rank early deaths better than late ones, falling from about 0.76 at one year to 0.72–0.73 at seven. XGBoost is ahead at every horizon, most clearly at one year (0.762 against 0.756 for the elastic-net), consistent with trees capturing some short-term effects such as the early harm of *NPM1* mutation. The fitted models gain most over the IPSS-R

at long horizons (0.051 for the elastic-net at seven years against 0.041 at one), because the IPSS-R was designed for myelodysplastic syndromes rather than for this mixed cohort.

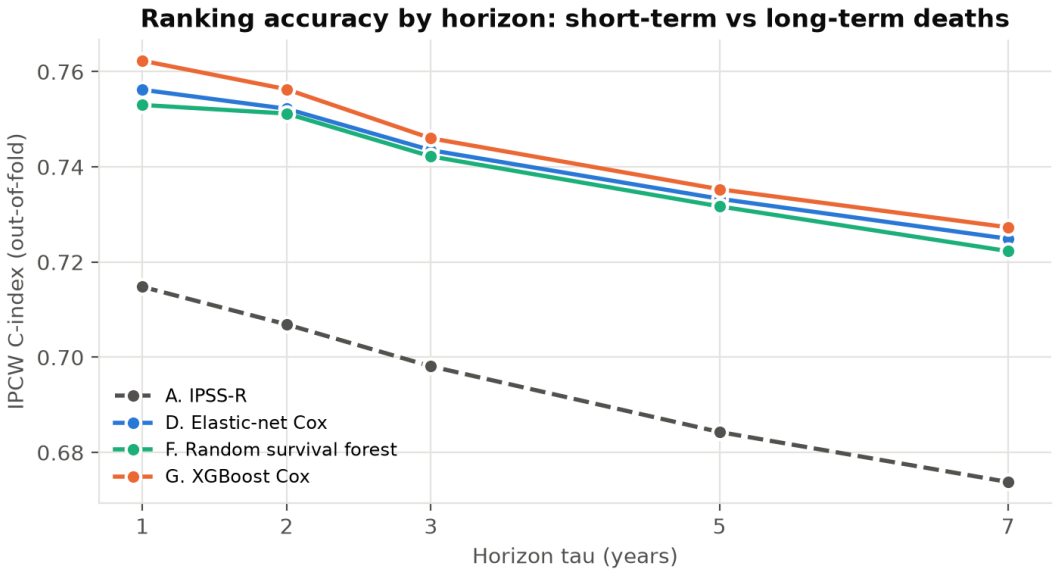


Figure 20: Out-of-fold IPCW C-index (repetition 0, mean of five folds) for horizons τ of one to seven years.

What the boosted trees use. Figure 21 shows each feature's share of the total improvement in the Cox loss ("gain"), summed over the five seed models of the final XGBoost. The IPSS-R score (20%) and the number of mutated genes (11%) dominate, followed by haemoglobin (8%), platelets (7%) and blasts (5%): the same picture as the elastic-net coefficients (Figure 17). Gain measures how much a model *uses* a feature, not its causal effect, and correlated features share the credit.

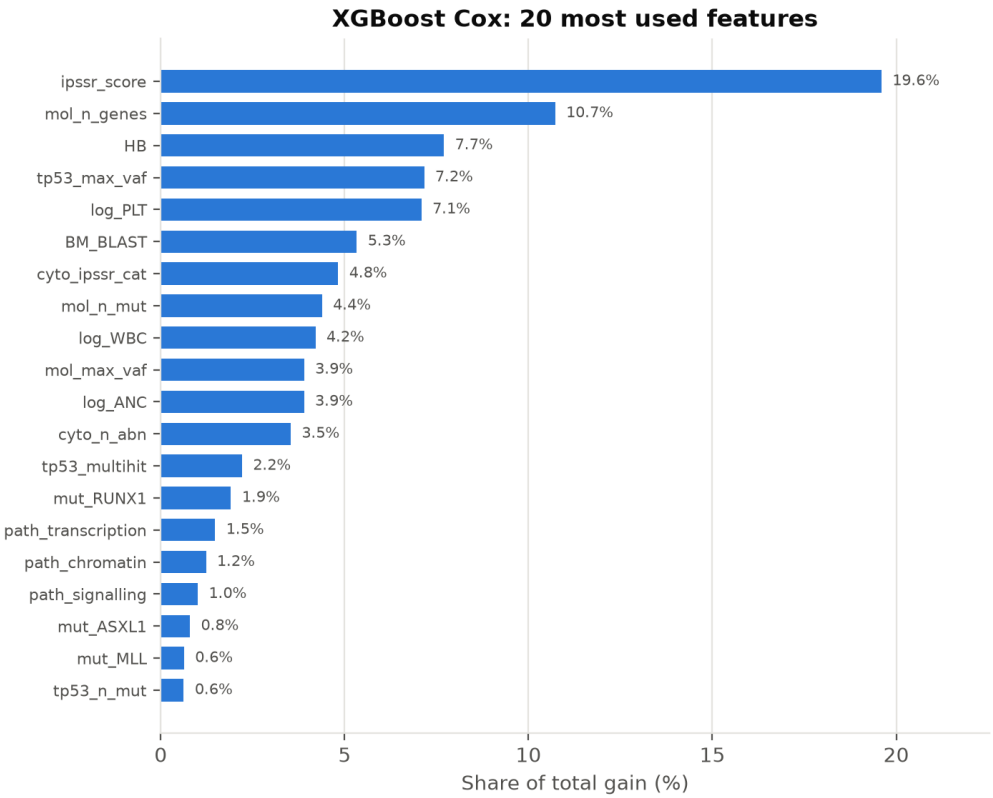


Figure 21: XGBoost Cox: the 20 features with the largest share of total gain.

9 Ensemble and interpretation (Stage 6)

9.1 Rank-averaged ensembles

The elastic-net, XGBoost and the forest produce risk scores on different scales: a linear predictor, a log relative risk and a summed cumulative hazard. Because the metric depends only on the order of the scores, each model's scores were converted to ranks between 0 and 1 within the set of patients being scored, and the ranks were averaged. For example, for four patients with elastic-net scores $(1.2, -0.3, 0.4, 2.0)$ and XGBoost scores $(0.9, 0.1, 0.8, 0.5)$, the ranks are $(0.75, 0.25, 0.50, 1.00)$ and $(1.00, 0.25, 0.75, 0.50)$, and the ensemble score is their average $(0.875, 0.25, 0.625, 0.75)$. The two models disagree about patients 1 and 4, and the average splits the difference. Converting one model's scores to ranks leaves its own C-index unchanged; averaging the ranks of two models produces a new order.

No model was refitted for this step. The out-of-fold predictions of each model, saved for all three repetitions, were combined inside each validation fold, so the ensembles are scored on exactly the same 15 folds and remain honest: every prediction still comes from a model that never saw that patient's centre.

Table 22: Ensembles and single models over the 15 centre-grouped folds; paired differences are with the elastic-net Cox.

Model	Mean	SD	Paired vs D	SD of paired	Folds won
D. Elastic-net Cox	0.7218	0.021	–	–	–
F. Random survival forest	0.7191	0.021	−0.0026	0.0043	4/15
G. XGBoost Cox (5 seeds)	0.7244	0.020	+0.0026	0.0023	14/15
E1. Elastic-net + XGBoost	**0.7250**	0.021	**+0.0032**	**0.0012**	**15/15**
E2. Elastic-net + XGBoost + forest	0.7245	0.021	+0.0027	0.0016	15/15

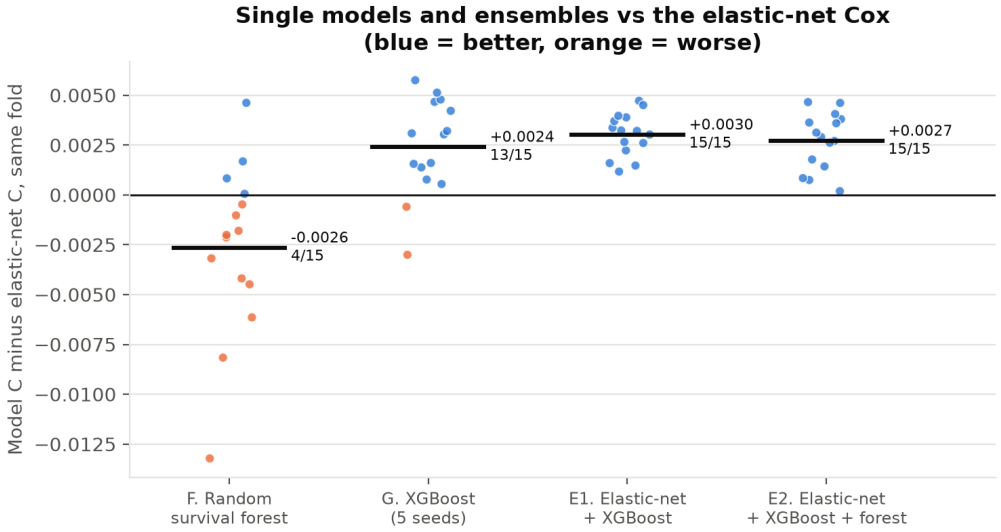


Figure 22: Paired difference with the elastic-net Cox on each of the 15 folds, for the two tree models and the two ensembles.

The two-model ensemble E1 is the best and, more importantly, the most *consistent* model (Table 22, Figure 22). It beats the elastic-net on all 15 folds, by +0.0032 on average, with a paired standard deviation of only 0.0012, so its mean gain is more than two and a half times its fold-to-fold spread. Against XGBoost alone the ensemble adds only +0.0006 on average; its value is mainly that it removes XGBoost's losing fold and narrows the spread. Adding the forest (E2) does not help. The gains are small because the models largely agree: inside the validation folds, the average Spearman correlation of the risk scores is 0.963 between the elastic-net and XGBoost, 0.966 between the elastic-net and the forest, and 0.964 between XGBoost and the forest. Averaging helps only where models disagree, and these models disagree on few patients.

9.2 Permutation importance

Gain (Figure 21) and coefficients (Figure 17) describe what a model uses on its training data. Permutation importance [17] instead measures what the model loses on *unseen* patients when a kind of information is scrambled. In each of the five folds of the first repetition, the elastic-net and XGBoost were fitted on the training part. Then, in the validation fold, the columns of one feature group were shuffled together among the patients (five times), and the drop in IPCW C-index was recorded. Features were shuffled in groups because related features (the 18 karyotype columns, for instance) carry overlapping information, and shuffling one alone would barely hurt while its partners still hold the information.

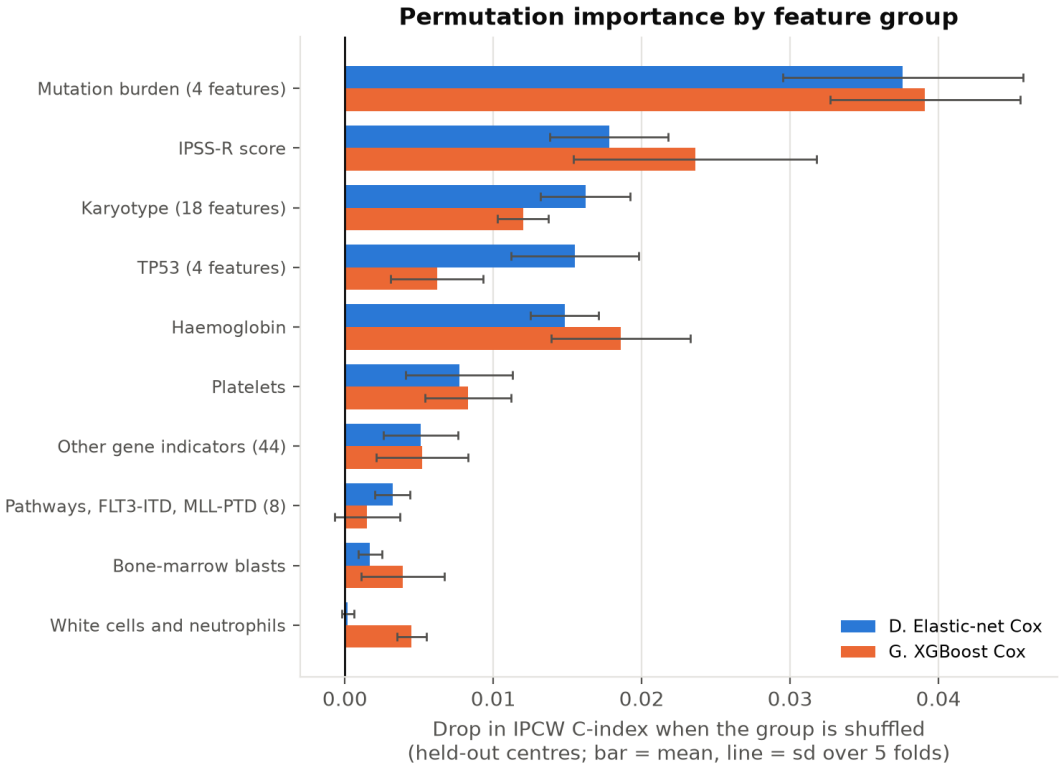


Figure 23: Drop in IPCW C-index on held-out centres when a feature group is shuffled (mean and standard deviation over five folds).

Table 23: Permutation importance: mean drop in IPCW C-index over five held-out folds.

Feature group	Elastic-net	XGBoost
Mutation burden (4 features)	0.038	0.041
IPSS-R score	0.018	0.024
Karyotype (18 features)	0.016	0.012
TP53 (4 features)	0.016	0.007
Haemoglobin	0.015	0.018
Platelets	0.008	0.008
Other gene indicators (44)	0.005	0.005
Pathways, FLT3-ITD, MLL-PTD (8)	0.003	0.001
Bone-marrow blasts	0.002	0.003
White cells and neutrophils	0.000	0.004

Three findings stand out (Figure 23, Table 23).

1. **Mutation burden matters most** for both models: the number of mutations and of mutated genes, the maximum VAF and the number of truncating mutations, all counted on the shared gene panel. Scrambling them costs about 0.04, more than twice any other group. This supports the decision in Stage 3 to count only genes sequenced in both centres: these counts are the models' strongest signal.

2. **The two models use TP53 differently.** The elastic-net loses 0.016 without the TP53 group, XGBoost only 0.007. The lesion frequencies of Stage 3 are separate prevalences and cannot show whether TP53 and complex karyotypes occur in the *same* patients, so

this was checked directly (Table 24). They do: 88.5% of TP53 multi-hit patients have a complex karyotype, against 4.7% of the others, and 60.3% of all complex karyotypes are TP53 multi-hit. A tree can build rules such as "complex karyotype *and* high VAF" that recover most of the TP53 signal when the TP53 columns are shuffled; a linear model adds effects and relies on the TP53 columns directly. The smaller drop for XGBoost means its TP53 group is less *irreplaceable*, not that TP53 matters less. TP53 multi-hit patients do not carry more mutated genes (a median of 2 in both groups), so the overlap runs through the karyotype and a higher VAF, not through mutation burden.

3. **Blasts look unimportant only because of overlap.** Bone-marrow blasts were the strongest single raw variable in Stage 2, yet shuffling them alone costs almost nothing. The IPSS-R score already contains the blast category, so the information survives in another group. Permutation importance measures what is *unique* to a group, not how prognostic it is on its own. Because the IPSS-R score also contains haemoglobin, platelets, neutrophils and cytogenetics, the drop from shuffling blasts and the IPSS-R score together cannot be found by adding their separate drops; it would need a joint permutation.

White cells and neutrophils are used by XGBoost (0.004) but not by the elastic-net, a hint of a non-linear effect that a straight-line Cox term cannot capture.

Table 24: TP53 multi-hit and karyotype in the same patients (training patients with a usable karyotype).

	TP53 multi-hit	Not multi-hit
Patients	209	2,614
Complex karyotype (%)	88.5	4.7
Monosomal karyotype (%)	73.7	3.0
Median mutated genes	2	2
Median maximum VAF	0.51	0.41

9.3 Final model choice

The final model was chosen by paired cross-validation, before its public score was known:

- **Final choice: E1, the rank average of the elastic-net Cox and the seed-averaged XGBoost Cox** (cross-validated IPCW C-index 0.725 ± 0.021). It has the highest mean, it wins all 15 folds against the elastic-net, and it has the smallest paired spread of any model.
- It combines an interpretable linear model, whose hazard ratios and coefficients are reported in Stage 4, with a tree model that can use non-linear and short-term effects.
- The improvement over the elastic-net alone is small (+0.003). The conclusions of this report do not depend on it: most of the predictive power comes from the feature engineering, which added +0.028 over the benchmark variables.

10 External validation on the test centre (Stage 7)

The public leaderboard scores each submission on part of the test centre's 1,193 patients, whose outcomes are never seen. It is therefore the one truly external check of this work: a different hospital, with a different case mix and gene panel. Submissions are limited to two per 24 hours. Before each of the last two uploads, the expected public score was written down. The prediction

added the model's cross-validated paired gain over the elastic-net to the elastic-net's known public score, for example $0.7543 + 0.0032 = 0.7575$ for the ensemble.

Table 25: Every submission: cross-validated score, prediction written before uploading, and public score.

Date	Model	CV	Predicted	Public	Public − CV
23 Oct 2026	Elastic-net Cox	0.7218	–	0.7543	+0.033
23 Oct 2026	IPSS-M-style Cox	0.7159	–	0.7374	+0.022
24 Oct 2026	**Ensemble (final)**	0.7250	0.7575	**0.7552**	+0.030
24 Oct 2026	XGBoost Cox (5 seeds)	0.7244	0.7569	0.7504	+0.026

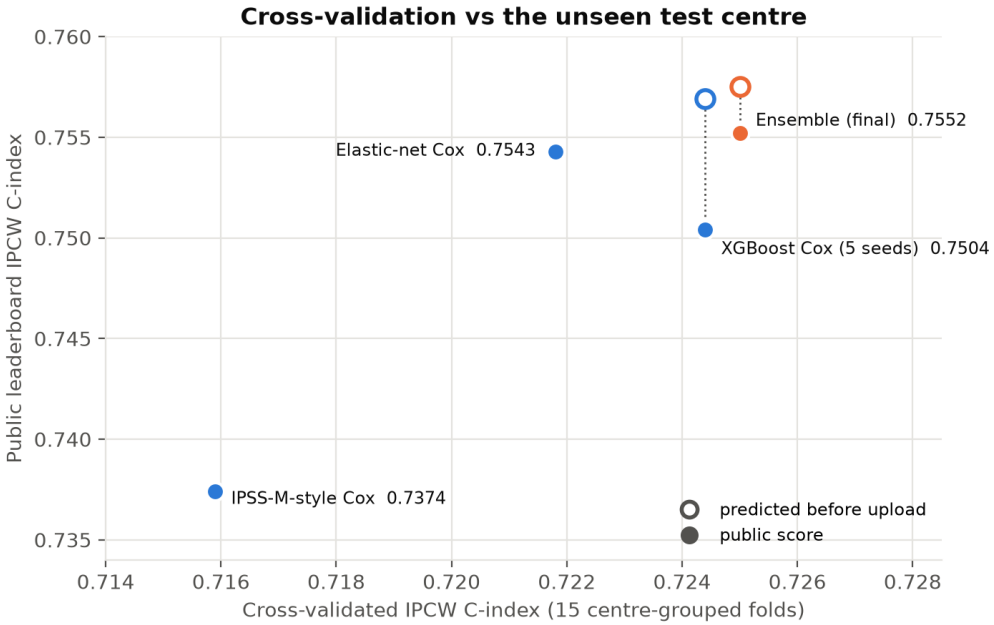


Figure 24: Cross-validated against public IPCW C-index. Hollow markers are the predictions written down before uploading; dotted lines join them to the actual scores.

The final model is the best on the test centre. The ensemble scores 0.7552, the highest public score of this work, within 0.0023 of its prediction (Table 25, Figure 24). Choosing it by cross-validation, without looking at its public score first, was confirmed.

XGBoost alone did worse than expected. Although it beat the elastic-net on 14 of 15 cross-validation folds, XGBoost scored 0.7504 on the test centre, 0.0039 *below* the elastic-net and 0.0065 below its prediction. The test centre behaved like one of its rare losing folds, and the ensemble, by averaging it with the elastic-net, still came out on top. This is exactly the protection that the small paired spread of the ensemble promised.

One natural explanation was tested and rejected. Tree models can be weak for patients with extreme values, and the test centre has many more very high-risk patients. If that were the cause, XGBoost should rank high-risk patients worse than the elastic-net in cross-validation too. Table 26 shows the opposite. Inside the IPSS-R very-high-risk subgroup, XGBoost ranks patients better than the elastic-net (0.623 against 0.608, winning 13 of 15 folds), and elsewhere the two are close. Two explanations remain untested. The first is chance: the public score uses only part of one centre, so a difference of 0.004 is within its sampling noise. The second is a measurement shift at the test centre: trees rely on sharp thresholds (for example, "haemoglobin

below 9.1"), so a small systematic difference in how a laboratory measures a value moves many patients across a threshold at once, whereas a linear model shifts its scores smoothly.

Table 26: XGBoost against the elastic-net inside subgroups (out-of-fold IPCW C-index, mean over folds with enough deaths to score).

Subgroup	Patients	Elastic-net	XGBoost	Difference	XGBoost wins
IPSS-R very high (> 6)	363	0.608	0.623	+0.014	13/15
IPSS-R high or very high (> 4.5)	822	0.635	0.642	+0.007	8/15
Blasts $\geq 10\%$	663	0.658	0.661	+0.004	6/15
Complex karyotype	307	0.676	0.675	−0.000	7/14
All others (IPSS-R ≤ 4.5)	2,351	0.689	0.691	+0.001	11/15

Public scores are higher than cross-validation. All four public scores exceed their cross-validated means, by 0.022 to 0.033. The C-index depends on the population as well as on the model: a population with a wider range of risk contains more pairs that are easy to order. The test centre has many very high-risk patients (a third are IPSS-R very high) alongside low-risk ones. Its censoring pattern and the size of the public subset also affect the score, so the case mix is a plausible explanation rather than a proven one. The offset is not the same for every model, and the order of models changes between cross-validation and the test centre. That is why models were chosen by paired cross-validation and the leaderboard was used only as a check.

11 Benchmarks, uncertainty and calibration (Stage 8)

This section adds what a journal reviewer would ask for: a comparison with the current clinical standard, confidence intervals, time-dependent discrimination, calibration, and a description of the cohort. The numbers in this section come from a verification run on Linux. Quantities involving XGBoost or the ensemble can differ from the Windows run of Sections 8–9 by about 0.0002 (Section 8), so for example XGBoost's paired gain over the elastic-net appears here as +0.002 rather than +0.003. No conclusion is affected.

11.1 The cohort

Table 27 describes both cohorts. Sex is not recorded in the data and was inferred from the karyotype (XX female, XY or loss of Y male); it is unknown for more test patients because many test karyotypes are recorded only as "Normal" or "complex".

Table 27: Characteristics of the training cohort (patients with a recorded outcome) and the test centre. Median (first–third quartile) or n (%).

Characteristic	Training (23 centres)	Test (centre KYW)
Patients	3,173	1,193
Sex (inferred): female / male / unknown	33.3 / 52.7 / 14.0%	17.1 / 45.8 / 37.1%
Deaths	1,600 (50.4)	not available
Median overall survival, years	3.15	not available
Median follow-up (reverse Kaplan–Meier), years	3.84	not available
Bone-marrow blasts, %	3.0 (1.0–8.0)	6.0 (2.0–17.0)
Haemoglobin, g/dL	9.7 (8.5–11.1)	8.4 (7.4–10.1)
Platelets, G/L	123 (65–230)	74 (37–143)
White blood cells, G/L	4.1 (2.7–6.7)	2.9 (1.9–5.3)
Neutrophils, G/L	2.00 (1.00–3.70)	1.30 (0.54–2.64)
Karyotype missing or unusable	350 (11.0)	120 (10.1)
Normal karyotype	1,643 (51.8)	410 (34.4)
Complex karyotype	307 (9.7)	336 (28.2)
IPSS-R cytogenetics poor or very poor	408 (12.9)	380 (31.9)
Mutated genes per patient (shared panel)	2 (1–4)	2 (1–3)
TP53 multi-hit (IPSS-M definition)	210 (6.6)	213 (17.9)
SF3B1 mutated	727 (22.9)	92 (7.7)
IPSS-R very high risk	363 (11.4)	395 (33.1)
IPSS-M very high risk	666 (21.0)	494 (41.4)

11.2 The IPSS-M as benchmark

The Molecular International Prognostic Scoring System (IPSS-M) [18] is the current clinical standard for myelodysplastic syndromes. It combines haemoglobin, platelets, blasts, the IPSS-R cytogenetic category, TP53 multi-hit status, MLL-PTD, FLT3 and 17 other genes with published weights:

$$\text{IPSS-M} = \frac{1}{\ln 2} \sum_{k} \beta_k \, (x_k - \mu_k) \,, \tag{8}$$

where β_k is the published coefficient of variable k and μ_k its reference mean. Blasts are capped at 20% and divided by 5, platelets capped at 250 G/L and divided by 100, and the residual genes enter as their count capped at 2. The weights were taken from the authors' open-source calculator [19]. A missing input receives its reference mean, so it contributes nothing; only published constants are used, and nothing is learned from these data. The implementation was checked against the official calculator on 300 training patients with complete data: all 300 scores agreed exactly. One deviation is unavoidable: copy-neutral loss of heterozygosity of TP53 is not available, so a single TP53 mutation counts as multi-hit only with VAF above 0.55 or a 17p loss.

An important caveat. The IPSS-M was developed on 2,957 patients of the International Working Group for Prognosis in MDS, an international consortium that very probably overlaps with this challenge's 23 training centres (the challenge is run with the institute of the IPSS-M's first author). If so, its weights were partly fitted to these patients, and its scores on the training data are partly in-sample and therefore optimistic. Its fair comparison is on the test centre.

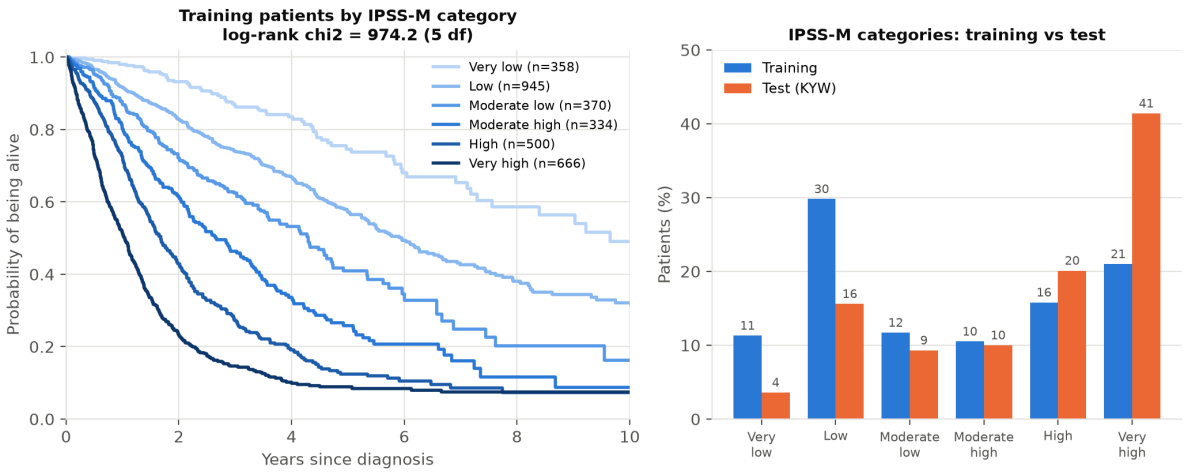


Figure 25: Left: survival by IPSS-M category in the training data. Right: share of patients in each category, training against test.

The six IPSS-M categories separate survival very clearly (log-rank $\chi^2 = 974.2$ on 5 degrees of freedom, against 737.4 on 4 for the IPSS-R), from a median of 9.65 years in the very low group to 1.03 years in the very high group (Figure 25). At the test centre, 41% of patients are IPSS-M very high risk, against 21% in training.

Table 28: The IPSS-M against fitted models on the same 15 centre-grouped folds (positive: the fitted model is better).

Comparison	Mean C	Paired difference	Folds won
IPSS-M score (no fitting)	0.720	–	–
IPSS-R score minus IPSS-M	0.669	−0.052	0/15
Elastic-net Cox minus IPSS-M	0.722	+0.002	10/15
XGBoost Cox minus IPSS-M	0.724	+0.004	10/15
Ensemble (final) minus IPSS-M	0.725	+0.005	11/15

Table 28 gives the central result of this section. With no fitting at all, the IPSS-M reaches 0.720, essentially the level of every model fitted in this work. The final ensemble is ahead by only 0.005 on average and wins 11 of 15 folds. Since the IPSS-M may itself be partly in-sample here, the comparison is uncertain in both directions.

11.3 Confidence intervals

The 15 cross-validation scores are not independent: the three repetitions reuse the same patients, and the training sets of different folds overlap. The naive standard error $s/\sqrt{J}$ (with $J = 15$ folds and s the standard deviation across folds) is therefore too small. The corrected resampled t-test of Nadeau and Bengio [20] inflates the variance to

$$\widehat{\mathrm{Var}} = \left(\frac{1}{J} + \frac{n_{\mathrm{valid}}}{n_{\mathrm{train}}}\right) s^2 = \left(\frac{1}{15} + 0.257\right) s^2 = 0.323\, s^2,$$

where 0.257 is the average ratio of validation to training patients. Since $\sqrt{0.323/0.067} = 2.2$, the corrected standard error is about 2.2 times the naive one. Intervals use the t distribution with 14 degrees of freedom ($t_{0.975} = 2.145$).

Table 29: Cross-validated IPCW C-index and paired differences with corrected 95% confidence intervals and two-sided p-values.

Quantity	Estimate	95% CI	p
IPSS-R	0.669	0.634 to 0.703	
IPSS-M	0.720	0.693 to 0.748	
Elastic-net Cox	0.722	0.696 to 0.747	
XGBoost Cox	0.724	0.699 to 0.749	
Ensemble (final)	0.725	0.700 to 0.750	
IPSS-M vs IPSS-R	+0.052	+0.037 to +0.066	< 0.001
IPSS-M-style Cox vs benchmark	+0.028	+0.010 to +0.045	0.004
Elastic-net vs IPSS-R	+0.053	+0.033 to +0.073	< 0.001
Elastic-net vs IPSS-M	+0.002	−0.007 to +0.011	0.71
XGBoost vs elastic-net	+0.002	−0.001 to +0.005	0.10
Ensemble vs elastic-net	+0.003	+0.002 to +0.004	< 0.001
Ensemble vs IPSS-M	+0.005	−0.004 to +0.013	0.28

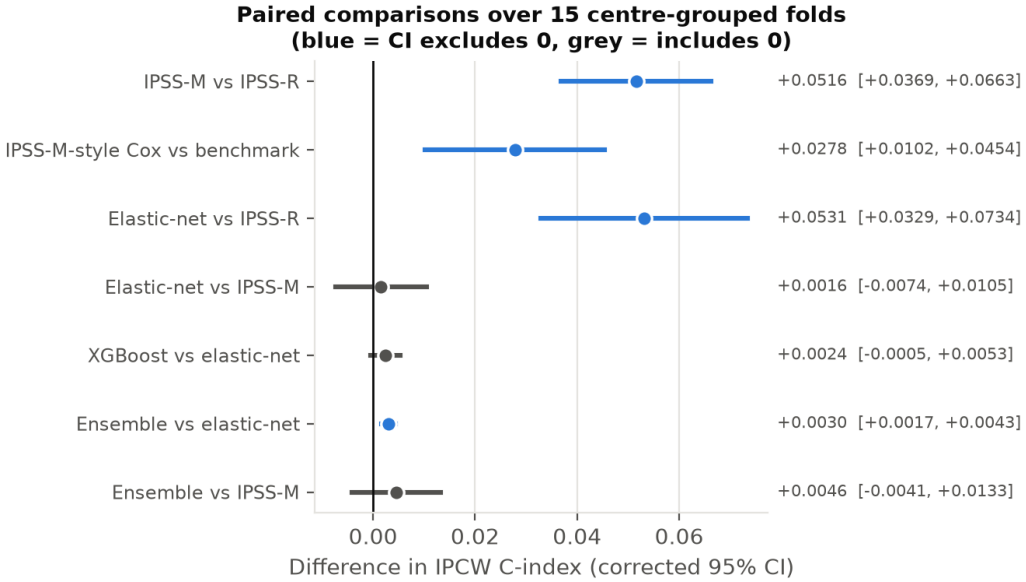


Figure 26: Paired differences in IPCW C-index with corrected 95% confidence intervals.

The intervals (Table 29, Figure 26) sort the comparisons clearly. The gains from clinical knowledge are large and certain: the IPSS-M and the fitted models beat the IPSS-R by about 0.05, and the literature-chosen features beat the benchmark variables by 0.028. The ensemble's gain over the elastic-net is tiny (0.003) but consistent enough to exclude zero. No fitted model is distinguishable from the IPSS-M: every interval includes zero.

11.4 Time-dependent AUC

The cumulative/dynamic AUC at time t [21] asks: among patients who died by year t and those still alive after it, how often does the score rank the death higher? It was computed on all 15 folds, with IPCW weights estimated on each training part. Validation follow-up was censored at 6.5 years, because some validation patients are followed for longer than anyone in the corresponding training part, where the censoring curve reaches zero; no reported time exceeds 6 years.

Table 30: Time-dependent AUC (mean over 15 folds).

Model	1 year	3 years	5 years
IPSS-R	0.723	0.742	0.725
IPSS-M	0.777	0.803	0.803
Elastic-net Cox	0.773	0.803	0.803
XGBoost Cox	0.779	0.802	0.804
Ensemble (final)	0.779	0.805	0.806

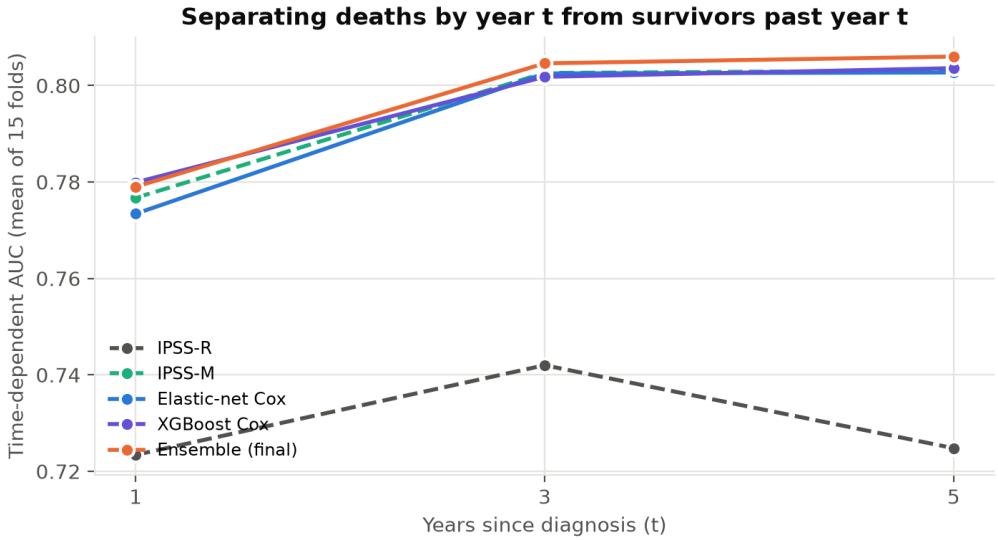


Figure 27: Time-dependent AUC at 1, 3 and 5 years (mean of 15 folds).

At every horizon the fitted models and the IPSS-M are within 0.006 of each other and about 0.05–0.08 above the IPSS-R (Table 30, Figure 27). The models discriminate better at three and five years (AUC about 0.80) than at one year (about 0.78).

11.5 Brier score and calibration

The C-index only judges the ranking. A clinician also needs predicted *probabilities* to be right: of 100 patients given a 60% chance of surviving three years, about 60 should survive. Three models giving survival probabilities were compared on the five folds of the first repetition: a Kaplan–Meier curve with no features (a reference), the IPSS-M recalibrated, and the elastic-net Cox with its Breslow baseline survival. "Recalibrated" means a one-variable Cox model on the IPSS-M score, fitted on each training part: the IPSS-M weights stay published, and only the link from score to probability is estimated. The Brier score [22] at time t is the average squared difference between the predicted survival probability and the observed status (alive or dead) at t, weighted by IPCW; the integrated Brier score averages it over 0.25 to 6 years. Lower is better.

Table 31: Brier scores (lower is better; mean of five folds) and mean absolute calibration error at three years over ten risk groups.

Model	Integrated Brier	Brier at 3 years	Calibration error at 3 years
Kaplan–Meier (no features)	0.211	0.243	–
IPSS-M (recalibrated)	**0.160**	**0.178**	**0.050**
Elastic-net Cox	0.163	0.181	0.065

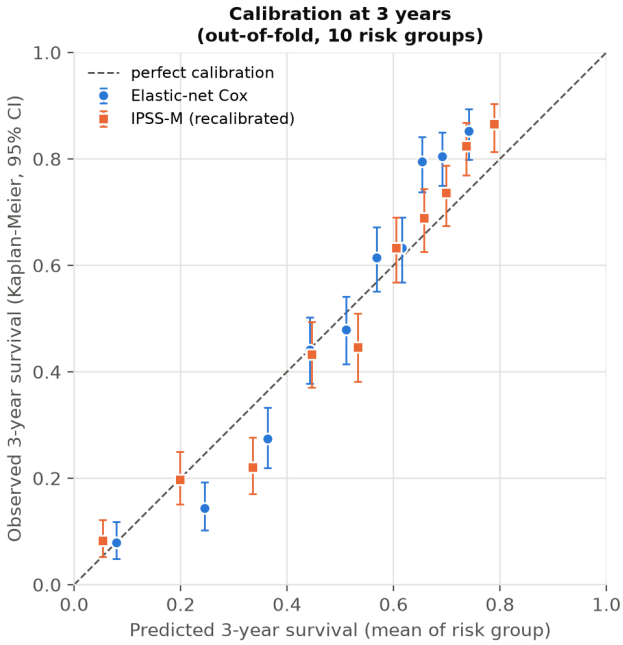


Figure 28: Calibration at three years: mean predicted against observed (Kaplan–Meier) survival in ten groups of out-of-fold predicted risk.

Both models reduce the Brier score by about a quarter compared with the no-feature reference (Table 31). The recalibrated IPSS-M is slightly better calibrated than the elastic-net (Figure 28). The elastic-net's predictions are compressed towards the middle: its predicted three-year survival spans 0.08 to 0.74, while the observed survival of its risk groups spans 0.08 to 0.85, so low-risk patients do better than predicted. Such compression is the expected effect of a penalty that shrinks the coefficients towards zero, and it could be corrected by recalibrating the model's linear predictor, as was done for the IPSS-M.

11.6 Summary of Stage 8

Against the IPSS-R, every fitted model is clearly and significantly better. Against the IPSS-M, a published score with no fitting, they are statistically indistinguishable in discrimination, and the IPSS-M is slightly better calibrated. Since the IPSS-M was very probably developed partly on these same patients, its scores here may be optimistic; the test-centre comparison (an upload of the IPSS-M score to the leaderboard) is the fairest available check.

12 Discussion

What mattered. The largest gain in this project came from the features, not from the algorithm. Moving from the four benchmark variables to 24 variables chosen from the clinical literature (IPSS-R cytogenetics, IPSS-M genes, TP53 multi-hit, MLL-PTD) added 0.028 in cross-validation and won 14 of 15 folds. Every later step, namely 84 penalised features, boosted trees and the ensemble, added at most 0.006 more. Permutation importance confirms the point: the strongest signal is the mutation burden, a simple count that only became comparable across centres once it was restricted to the shared gene panel. The published IPSS-M, which encodes the same expert knowledge in fixed weights, performs as well as every fitted model here (Section 11). The honest conclusion is that this work reproduces the performance of the current clinical standard from the challenge data, with transparent and validated code, but does not surpass it.

Validation that matches the test. The test set is a hospital never seen in training, so every model was validated on held-out hospitals and compared fold by fold on the same 15 folds. The paired comparison removes the large differences in difficulty between folds (the IPSS-R score alone ranges from 0.624 to 0.729 across folds) and leaves each model's own advantage. Hyperparameters were always chosen by an inner, centre-grouped split, so no validation fold was used twice.

Checking the details. Three problems were found only by looking closely at intermediate results, and each would otherwise have biased the conclusions. The three "random" repetitions of cross-validation were identical, visible as every fold score appearing three times. A tuning grid was too narrow, with the smallest value chosen most often. And XGBoost gave different results on Windows and Linux. A fourth case was a hypothesis rather than a bug: the explanation that trees fail on high-risk patients was tested on the cross-validation predictions and rejected.

Clinical sense. The fitted hazard ratios agree with established knowledge: TP53 multi-hit (HR 3.21), MLL partial tandem duplication (2.05) and FLT3 mutation (1.65) carry the highest risk, and haemoglobin and platelets are protective. Two findings show why adjusted and unadjusted analyses differ. *SF3B1* is protective on its own but neutral once blasts, karyotype and blood counts are included, because *SF3B1* patients are favoured on most of those. And *NPM1* is harmful in the first months but protective later, a violation of proportional hazards that its single Cox coefficient hides.

13 Limitations

- **Overlap with the IPSS-M cohort.** The training data very probably overlap with the cohort on which the IPSS-M was developed. Its cross-validated scores here may be optimistic, and the data are not new to the field.
- **One external centre.** The test set comes from a single hospital, and the public score uses only part of it. Conclusions about how the models transfer rest on this one centre; the private leaderboard, updated on 15 June and 15 December, will give a second, larger look at the same centre.
- **Case-mix shift.** The test centre is much sicker than the training centres. The models were not adapted to it, for example by reweighting training patients to resemble the test centre.
- **Measurement differences.** The gene panels differ between centres, so 60 training genes and 598 training mutations were not used. Monocyte counts were dropped because they are mostly missing at the test centre. Laboratory calibration differences cannot be checked from the data.
- **The karyotype parser** is a set of hand-written rules tested on 25 cases and cross-checked against a simpler rule. It is not a validated ISCN parser, and rare or badly written karyotypes may be misread.
- **Censoring.** The metric and the Cox models assume that censoring is unrelated to the patient's prognosis. Follow-up practices that differ between centres could violate this.
- **Missing information.** There is no information on treatment (for example, stem cell transplantation), age, or disease subtype, all of which affect survival.
- **The benchmark model** was reconstructed from the challenge description, since the benchmark notebook was not available.

- **Computation.** XGBoost results differ in the fourth decimal place between operating systems; all reported numbers come from one machine.

14 Conclusion

For adult myeloid neoplasms, a carefully engineered set of clinical, cytogenetic and molecular features, validated on held-out hospitals, predicts overall survival with an IPCW C-index of about 0.72 in cross-validation and 0.755 on an unseen test centre, about 0.05 better than the IPSS-R clinical score. The final model, a rank-averaged ensemble of an elastic-net Cox and a gradient-boosted Cox model, was chosen by paired cross-validation before its public score was known, and it was the best submitted model on the test centre. It does not, however, measurably improve on the published IPSS-M, which reaches the same discrimination with no fitting and slightly better calibration. The main lessons are that domain knowledge and a validation design that mirrors the real test mattered far more than the choice of algorithm, and that a new model should always be compared with the strongest existing clinical score before any claim of improvement.

References

[1] D. P. Noren *et al.*, A crowdsourcing approach to developing and assessing prediction algorithms for AML prognosis, *PLOS Computational Biology*, 12(6): e1004890, 2016.

[2] F. E. Harrell, R. M. Califf, D. B. Pryor, K. L. Lee and R. A. Rosati, Evaluating the yield of medical tests, *JAMA*, 247(18): 2543–2546, 1982.

[3] H. Uno, T. Cai, M. J. Pencina, R. B. D'Agostino and L. J. Wei, On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data, *Statistics in Medicine*, 30(10): 1105–1117, 2011.

[4] S. Pölsterl, scikit-survival: a library for time-to-event analysis built on top of scikit-learn, *Journal of Machine Learning Research*, 21(212): 1–6, 2020.

[5] E. L. Kaplan and P. Meier, Nonparametric estimation from incomplete observations, *Journal of the American Statistical Association*, 53(282): 457–481, 1958.

[6] N. Mantel, Evaluation of survival data and two new rank order statistics arising in its consideration, *Cancer Chemotherapy Reports*, 50(3): 163–170, 1966.

[7] C. Davidson-Pilon, lifelines: survival analysis in Python, *Journal of Open Source Software*, 4(40): 1317, 2019.

[8] P. L. Greenberg *et al.*, Revised International Prognostic Scoring System for myelodysplastic syndromes, *Blood*, 120(12): 2454–2465, 2012.

[9] J. Schanz *et al.*, New comprehensive cytogenetic scoring system for primary myelodysplastic syndromes (MDS) and oligoblastic acute myeloid leukemia after MDS derived from an international database merge, *Journal of Clinical Oncology*, 30(8): 820–829, 2012.

[10] D. R. Cox, Regression models and life-tables, *Journal of the Royal Statistical Society, Series B*, 34(2): 187–220, 1972.

[11] H. Zou and T. Hastie, Regularization and variable selection via the elastic net, *Journal of the Royal Statistical Society, Series B*, 67(2): 301–320, 2005.

[12] N. Simon, J. Friedman, T. Hastie and R. Tibshirani, Regularization paths for Cox's proportional hazards model via coordinate descent, *Journal of Statistical Software*, 39(5): 1–13, 2011.

[13] P. M. Grambsch and T. M. Therneau, Proportional hazards tests and diagnostics based on weighted residuals, *Biometrika*, 81(3): 515–526, 1994.

[14] H. Ishwaran, U. B. Kogalur, E. H. Blackstone and M. S. Lauer, Random survival forests, *The Annals of Applied Statistics*, 2(3): 841–860, 2008.

[15] J. H. Friedman, Greedy function approximation: a gradient boosting machine, *The Annals of Statistics*, 29(5): 1189–1232, 2001.

[16] T. Chen and C. Guestrin, XGBoost: a scalable tree boosting system, *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794, 2016.

[17] L. Breiman, Random forests, *Machine Learning*, 45(1): 5–32, 2001.

[18] E. Bernard *et al.*, Molecular International Prognostic Scoring System for myelodysplastic syndromes, *NEJM Evidence*, 1(7): EVIDoa2200008, 2022.

[19] Papaemmanuil Lab, `ipssm`: IPSS-M risk calculator, JavaScript package version 1.0.4 (MIT licence), https://www.npmjs.com/package/ipssm, 2023.

[20] C. Nadeau and Y. Bengio, Inference for the generalization error, *Machine Learning*, 52(3): 239–281, 2003.

[21] H. Hung and C.-T. Chiang, Estimation methods for time-dependent AUC models with survival data, *Canadian Journal of Statistics*, 38(1): 8–26, 2010.

[22] E. Graf, C. Schmoor, W. Sauerbrei and M. Schumacher, Assessment and comparison of prognostic classification schemes for survival data, *Statistics in Medicine*, 18(17–18): 2529–2545, 1999.